

A-2000wf

AUTO-ISOKINETIC FLOW INTEGRATOR

USER GUIDE

Version wf-v9

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1 INTRODUCTION

The Environmental Supply Company's A-2000 Auto-Isokinetic Flow Integrator is engineered for the automation of isokinetic source sampling procedures. The A-2000 performs under normal operating conditions encountered by stack sampling professionals to satisfy the requirements as specified by the Environmental Protection Agency in the Office of Air Programs Publication No. APTD-0576. The system employs an array of features designed in response to the suggestions and requests of stack sampling professionals for accuracy, reliability, ease of use in operation, maintenance and calibration.

1.1 OPERATIONAL OVERVIEW OF THE SYSTEM

Overview (ESC Automated Isokinetic System)

1. The A-2000 Auto-Isokinetic Flow Integrator System uses the same manual system components from the nozzle to the impinger box as described in US EPA Method 5.
2. The cool dry sample leaving the last impinger passes through a stainless steel umbilical adapter that contains a thermocouple (TC#4) and a calibrated Venturi used to measure the sample flow rate. The TC measures the dryer exhaust temperature as required for US EPA Method 5 and is also used as the temperature of the Venturi in calculating the sample flow rate. The accumulated sample volume is calculated by integrating the Venturi flow rate over time.
3. The auto-positioning valve in the A-2000 Auto-Isokinetic Flow Integrator continuously controls the sample flow.
4. The sample gas is extracted from the A-2000 Auto-Isokinetic Flow Integrator using an external vacuum pump. This may be a stand-alone vacuum pump (not required to be leak free) or a typical US EPA Method 5 manual control console with pump.



Shown on left:

The flow measuring Venturi is an integral part of the typical Cold Box Umbilical

2 AUTO-ISOKINETIC OPERATING INSTRUCTIONS

2.1 GETTING READY

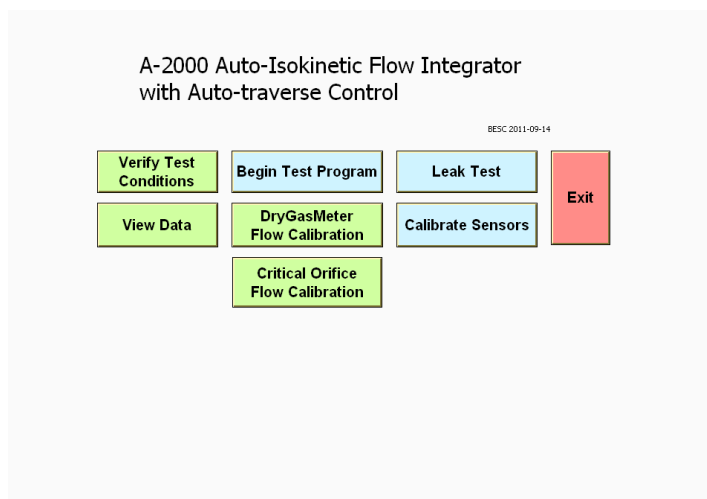
Before going into the field, all components of the test equipment should be checked individually and calibrated if necessary. All of the normal Method 5 preparations should be made such as probes, filters, nozzles, and impingers. In addition you should become familiar with the operation of all of the applicable hardware and the performance of the software program so that you can simulate a successful stack test in the lab.

Once all the test equipment is in place, a typical test program might proceed in the following order:

1. Start the A-2000 Program.
2. Run 'View Data' to see if sensor data is being transmitted properly.
3. Run 'Calibrate Sensors' to check or calibrate as needed.
4. Run 'Dry Gas meter Flow Calibration' as needed.
5. Run 'Critical Orifice Calibration' as needed.
6. Run 'View Data' to check communications or to determine nozzle size and flow settings if needed.
7. Run 'Verify Test Conditions' to enter nozzle size, run times, and other test conditions.
8. Run 'Leak Test' for the recommended pre-leak test.
9. Position probe at test point 1.
10. Run 'Begin Test Program' to begin sampling and logging data.
11. Pause program to change filter or perform other necessary tasks.
12. Continue program to completion and final data log.
13. Run 'Leak Test' for the post-leak test.
14. Run 'Calibrate Sensors' for post-test calibration check.

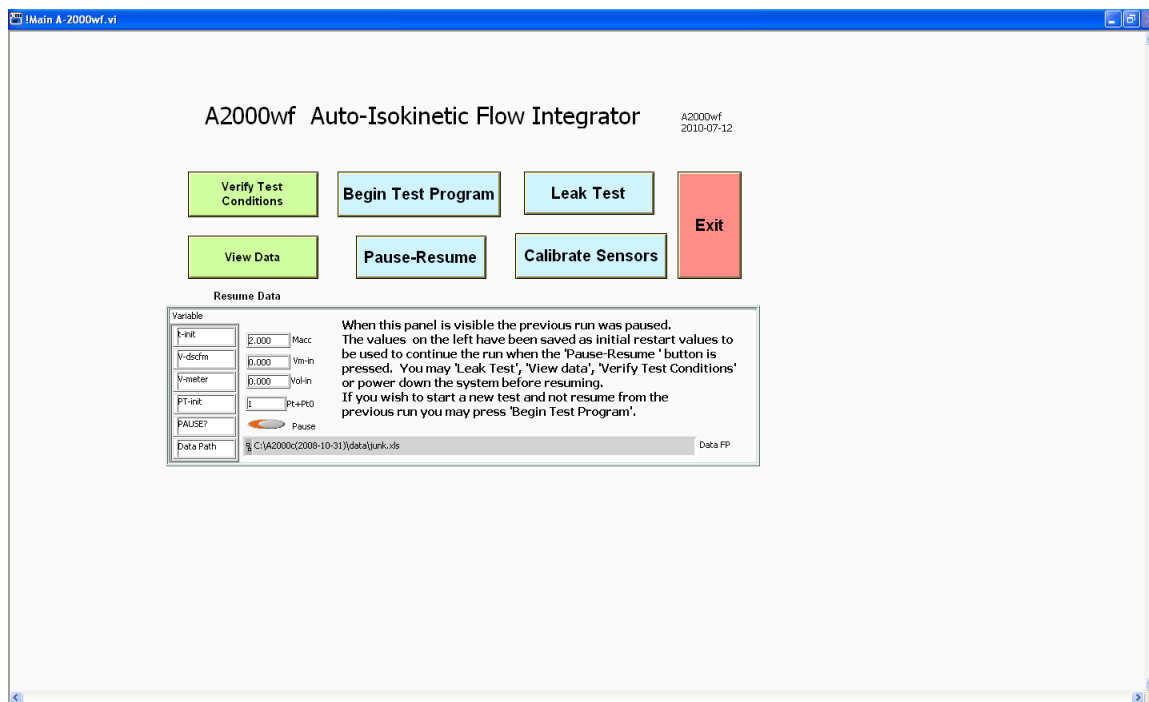
2.2 STARTING THE PROGRAM

To launch the program, click on the A2000wfxxxx program icon on your computer desktop or from your program file directory. The main menu panel will appear. The labeled buttons take you through the appropriate program steps.



2.2.1 PAUSING THE PROGRAM

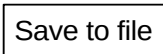
If the test needs to be paused for changing filters or any other reason where you want to continue sampling later at the same place in the stack, the 'Pause-Resume' button appears. If you close the program and turn off the PC you can later return and continue the test at the place where it was paused. If you decide you don't want to continue the run, you can ignore the flashing 'Pause-Resume' button and start a new run with 'Begin Test Program'.



2.3 VERIFY TEST CONDITIONS

Click on the 'Verify Test Conditions' button to display the test conditions input and verification panel that displays data that is read from the C:\Program Files\A2000wf\config\current.cfg file in the 'config' directory.

- | | |
|-----------------------------|---|
| Read from file | Data from a previously saved file may be entered using the 'Read from file' button. This data is reviewed and new data can be entered for the current test. |
| Save to file | Current test condition values can be saved using the 'Save to file' button. The operator will be prompted for a new configuration file name. Caution!---If the file name currently exists then the file will be updated with no warning that a file is being overwritten. The 'Save to file' button is also a good method of making a backup of the current configuration. |
| Use current settings | If the initial data is altered or a different data file is loaded then the 'Use current settings' button must be pressed which causes the new data to be written to the 'current.cfg' file and then returns to the main menu. |
| Cancel changes | The 'Cancel changes' button returns to the main menu with the initial data unchanged. |

A backup of this data should be made using the  button. When entering this routine, the initial data is the data from the previous entry/exit and any calibration changes that have been made in the 'Calibrate Sensors' program.

2.3.1 ΔP , ANALOG GAIN AND OFFSET

The screenshot shows a software window titled 'dP, Analog Gain and Offset' with tabs for 'dP, Analog Gain and Offset', 'Cp and Flow Parameters', 'Stack Conditions', and 'Data Acquisition'. The main area is titled 'Transducer Gain and Offset' and contains a table with two rows of data for different transducers.

	Pitot	Static	Venturi	Barometer	Vacuum	NA6	NA7	NA8
dP, An.Gain	25.654	257.937	51.943	75.006	12900.762	1.000	1.000	0.000
dP, An.Offset	-1.219	-647.217	-5.029	-442.544	-254.635	0.000	0.000	0.000

Below the table are four buttons: 'Read from file', 'Save to file', 'Use current settings', and 'Cancel changes'. A note at the bottom states: 'Use 'Save to file' to back up all configuration settings'.

Clicking the 'dP, Analog Gain and Offset' tab allows you to view or change (not usually done here) the gain and offset from this calibration.

The use of electronic transducers in US EPA Method 5 requires that they be properly calibrated. This calibration may be done according to the procedure outlined in Method 2F.

The software module called 'Sensor Calibration', located on the main menu, can be used to expedite the calibration and writes the results to the C:\Program Files\A2000wf\config\current.cfg file in the 'config' directory where they can be used by the program.

The relation is:

$$\text{Value} = \text{gain} * (\text{V or I}) + \text{Offset}$$

2.3.2 Cp and Flow Parameters

The screenshot shows a software window titled 'Cp and Flow Parameters'. It contains a 'Parameter Input' section with the following fields and values:

Pitot	Venturi k	Venturi n	Nozzle Diameter (mm)	Optional DGM coefficient (Gamma)
Cp	k	n	Diam	Gamma
0.8400	26.6710	0.5000	3.8000	1.0000

Below the input fields are four buttons: 'Read from file', 'Save to file', 'Use current settings', and 'Cancel changes'.

The parameters C_p , Gamma, and Diam are US EPA Method 5 values for the current Pitot coefficient, dry gas meter efficiency and nozzle diameter, respectively. Gamma is only used when a gas meter is used for calibration or as a volume backup. The Venturi coefficients k and n are used to calculate the flow rate (Q) through the Venturi using the following equation:

$$Q = k * \sqrt{\Delta P}^n * \sqrt{\frac{T}{P * M}}$$

2.3.3 Stack Conditions

The screenshot shows the 'Stack Conditions' tab in a software application. The interface includes a menu bar (File, Edit, Operate, Tools, Window, Help) and a tabbed control area with four tabs: 'dP, Analog Gain and Offset', 'Cp and Flow Parameters', 'Stack Conditions' (selected), and 'Data Acquisition'.

Under the 'Stack Conditions' tab, there is a 'Stack Test Conditions' section with three input fields: '% H2O Estimate' (value: 1.000), '% CO2 Estimate' (value: 0.000), and '% O2 Estimate' (value: 20.800). Below these are three buttons: 'Read from file', 'Save to file', and 'Use current settings'.

The 'Standard Conditions' section is located below the test conditions. It includes 'Temperature' (Tstd, value: 20.000) and 'Pressure' (Pstd, value: 760.000) fields. There are also unit selection buttons for 'Deg C' and 'mmHg'.

The 'Port Traverse' section is on the right side of the 'Standard Conditions' section. It includes 'ID' (Port #, value: 1.000), 'Pts per port' (# of Pts, value: 5.000), and 'Time, Min each Pt' (Time-Pt, value: 5.000) fields. Below these fields are two buttons: 'Use current settings' and 'Cancel changes'.

The 'Stack Conditions' tab will usually require new Stack Test Conditions, Standard Conditions, and Port Traverse data with every test or at each port change.

Stack Test Conditions consist of preliminary percentage estimates of H₂O, Dry CO₂, and Dry O₂.

Standard Conditions are the Reference Temperature and pressure for correcting gas measurements to desired conditions.

Port Traverse data is the number of traverse points per port and the time at each point sets the total sample time at each port.

2.3.4 DATA ACQUISITION PARAMETERS

dP, Analog Gain and Offset Cp and Flow Parameters Stack Conditions **Data Acquisition** AutoTraverse

Data Acquisition Timing SW VERSION GOES HERE

Units ? Avg #'s for Screen update DAQ Sample Time, msec Desired Print Time, Min

English 10 200 0.500

Screen update time = (Avg # for Screen update) * (DAQ Sample Time)
Example: 10 x 200 = 2000 msec = 2 seconds

Read from file Save to file Use current settings Cancel changes

Use 'Save to file' to back up all configuration settings

The **'Units?'** button determines whether English or Metric units are to be used with the system. Typically it will not be changed after the first setup. Default configuration files for both units are saved in the C:\Program Files\A2000wf\config\ directory. It is a good idea to use the 'Save to file' button to make copies of the current.cfg file under a different name for later use.

'Avg #'s for Screen update' field is the number of readings taken, averaged and output to the screen display. An average number of 10 is recommended.

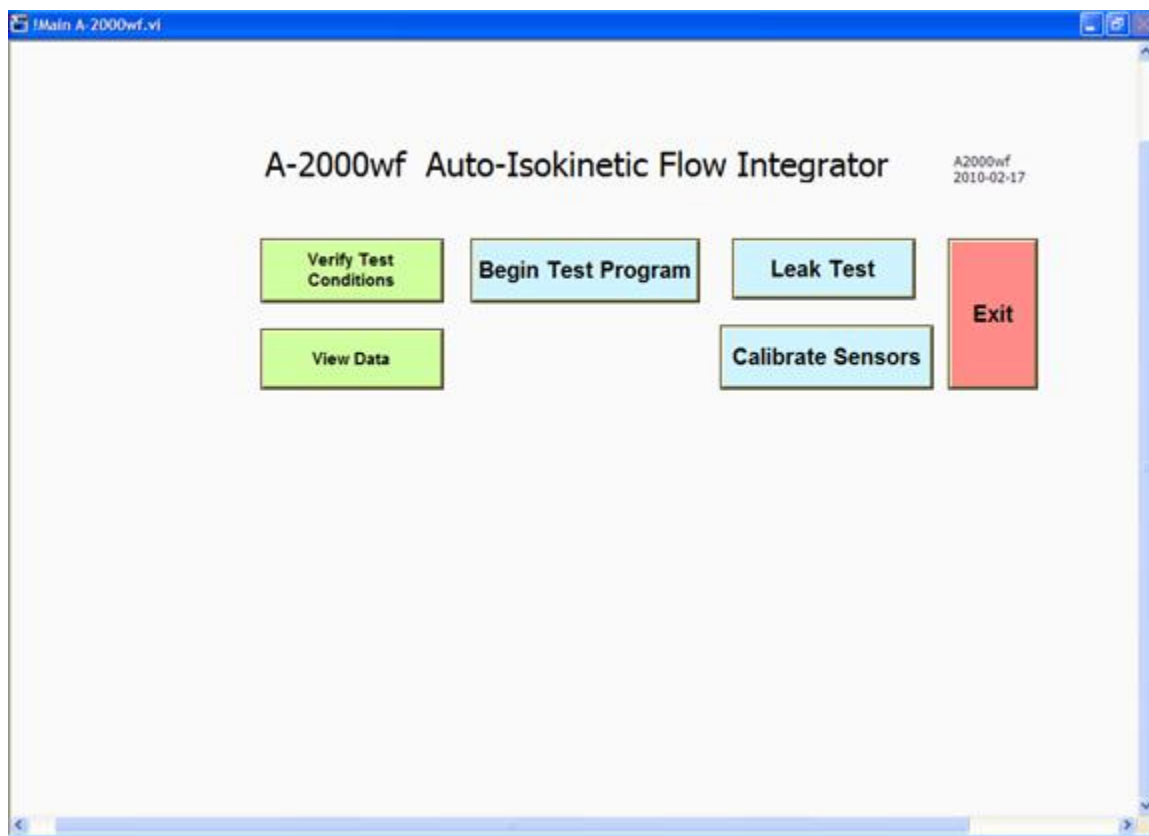
The **'DAQ Sample Time' is in milliseconds (msec)** and is the time allowed to take transducer readings and perform calculations. A value of 200 msec is recommended.

The product of the 'Avg #'s for Screen update' and 'DAQ Sample Time' values equals the time between screen updates. 10 readings times 200 ms DAQ Sample Time equals 2000 msec or 2 seconds.

The **'Desired Print Time'** is the time interval between data log entries. Data is averaged and written to the spreadsheet at the interval known as the Desired Print Time and the value is displayed in minutes. 0.500 minutes means data is time stamped and written to the spreadsheet every 30 seconds. The averages for the traverse point are also logged.

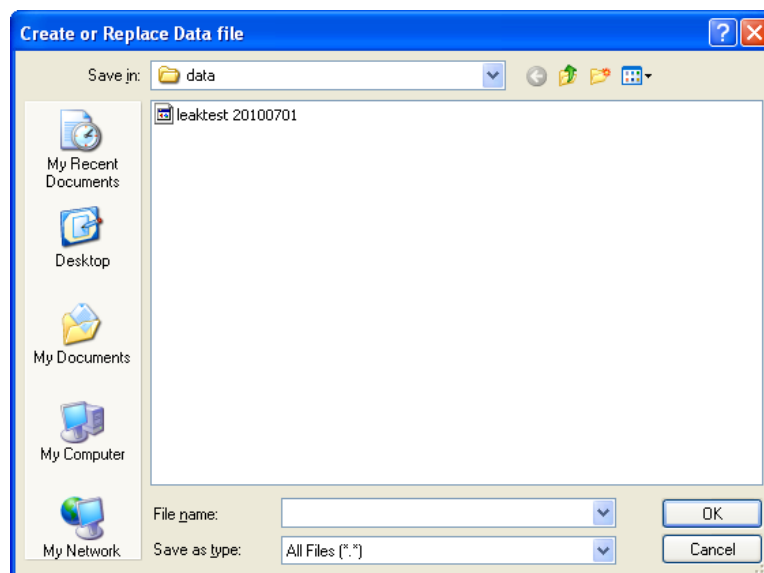
When the information in all four tabs is verified it may be saved to a file name of your choice. This is done with the 'Save to file' button. When you are ready to return to the main program, click on the 'Use current settings' button. This action stores the information in a file called 'C:\Program Files\A2000wf\config\current.cfg' for use by the main program.

2.4 LEAK TEST PROGRAM

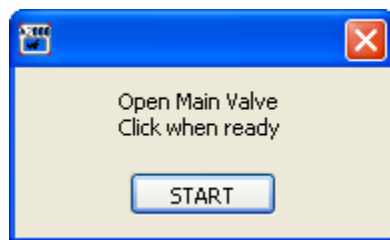


For the Pre-test leak check the sample train is fully set up for running the stack test but the probe is not yet inserted into the port. A suitable stopper is placed over the nozzle tip before the leak test begins. When the leak test is completed it is important to release the vacuum by removing the stopper on the nozzle or fluid may be sucked from some of the impinger bottles

1. On the front panel of the A-2000 Console place both the 'PUMP' switch and 'VALVE' switch to the left or the 'AUTO' position.
2. Click on the 'Leak Test' button to begin the leak test.
3. Enter a descriptive file name to save the data and click OK



4. You will be prompted to open the Venturi valve. When you are sure that the nozzle is capped and the Venturi valve is open click 'START' and the pump will begin evacuating the system. The rate of evacuation is limited by the control system.



5. Adjust the estimated volume of the impingers and the probe system if necessary.
6. Set the vacuum level desired for the test.
7. When the vacuum reaches the desired level and the pump shuts off, close the Venturi valve.

The leak rate can be observed on the panel. Data for the test is logged to the file chosen at the start of the run. Data is logged every 0.25 minutes during the leak test. After a sufficient time the test can be terminated with the 'Terminate Leak Test' button. The data sheet will indicate pass if the leak rate is less than 0.02 CFM.

Leak Test

Terminate Leak Test

Computations
pid control
Misc/error

Enter Volume of Impingers and Test Vacuum Here

Estimated Volume M

Liters

Leak Rate

LPM

Test vacuum M

mm Hg

Actual Vacuum

A2000wf
2010-07-10

Minutes Remaining

Leak test time defaults to 10 Min. It may be manually terminated.
Print is set for 0.25 Min.

NOTE: The 'pid control' and 'Misc/error' panels are primarily technical details that are not normally used during isokinetic sample collection.

Leak Test

Terminate Leak Test

Computations
pid control
Misc/error

Avg #'s

Print Time,min

Ven-vac gain

Ven-vac offset

Units ?

Sample Time,msec

dP,RH Gain

dP,RH Offset

dP,RH Offset

Avg dP@RH

0.650	0.250
1.000	0.000
0.500	20.800
0.000	68.000
-0.700	29.920
29.500	1.000
0.000	3.000
0.000	1.000

Tstd

Pstd

Minutes

A2000wf
2010-07-15

error Pt avg

error dP@RH

file path (dialog if empty)

.xls file preferred

Minutes Remaining

Leak test time defaults to 10 Minutes. It may be manually terminated.
Print is set for 0.25 Minutes.

2.5 BEGIN TEST PROGRAM

When all test setup is complete and you are ready to collect sample at traverse point '1', click on 'Begin Test Program'. You will be prompted to either create a new data or replace an existing file. Choose a name that is unique and descriptive of the job site and test being run. The last prompt will ask that the pump be turned on, the main valve opened and the Dry Gas Meter reading recorded.

The following panels will be available for viewing:

1. The 'Computations' panel is the primary panel of interest.
2. The 'Data' panel is helpful to check values of temperatures and pressures.

All the data on these two panels are written to the data directory on the computer. Other panels are usually not accessed during a test run.

Below all panels there are three status indicators. The 'Pause' button allows you to stop sampling for an indefinite period and resume when ready. The sample point indicator shows the traverse point where the probe should be located. The other indicators show the time remaining at that point and serve as a pre-warning in hundredth minutes before moving the probe to the next point.

The 'Terminate' button is used only when you want to terminate a traverse before completing all of the sample points (e.g. an aborted run).

2.5.1 Computations

M5 Sample Run Terminate Test

Computations Data Test Conditions pid controls Misc/error

Computations M

H ₂ O% Vol	1.000	% Est	
Wet Mol Wt	28.724		
Vel, Stack	4.420	M/sec	
Vel, Nozzle	0.000	M/sec	
Isokinetic	0.000	%	
Q _v	0.000	DALPM	Flow Rate at Venturi
Q _{dslpm}	0.000	DSLPM	Flow Rate at dry standard conditions
Vol (DGM)	0.000	DALiter	Volume at (optional)DGM conditions
Volume, dsliters	0.000	DSLiter	Volume at dry standard conditions

PAUSE Minutes Remaining: 4.87 at Sample Pt: 1 Probe should be at this location

H₂O% Vol is the moisture estimate originally entered.

Wet Mol Wt is calculated.

Vel, Stack is calculated from Pitot Transducer value.

Vel, Nozzle is calculated from Venturi Transducer value.

Isokinetic is controlled to value set in 'pid controls' (1.00 equals 100%)

Q_v is the flow rate at the Venturi in dry actual CFM (or LPM).

Q_{dslpm} is the conversion to dry standard conditions, i.e., T_{std} and P_{std} that were chosen in 'Verify Test Conditions'.

Vol (DGM) is the conversion of this flow rate to conditions of an optional (backup) dry gas meter with its temperature measured at TC#6.

Volume, dsliters is the flow rate integrated over time.

Equations for calculations are shown in the appendix.

2.5.2 Data

The screenshot shows the 'M5 Sample Run' window with the 'Data' tab selected. The window has a menu bar (File, Edit, Operate, Tools, Window, Help) and a toolbar with a 'Terminate Test' button. The main area is divided into two sections: 'Analog Data' and 'Temperature'. The 'Analog Data' section contains a table of sensor readings. The 'Temperature' section contains a table of temperature readings. Below these tables is a 'PAUSE' button and a 'Minutes Remaining' field showing 4.64. The 'Sample Pt' is set to 1. A note indicates the probe should be at a specific location.

M5 Sample Run Terminate Test

Computations **Data** Test Conditions pid controls Misc/error

Analog Data

Pitot	1.690	mmH2O
Static	6.727	mmHg
Venturi	0.000	mmH2O
Barometer	742.355	mmH2O
Vacuum	1.511	mmHg
NA6	0.000	
NA7	0.000	
Ven/Vac	0.000	

Temperature deg C

Stack	20.6
Probe	21.0
Filter Box	21.3
Dryer	1770.0
Filter Out	1770.0
DGM	21.3
aux	1770.0
Console	21.9

PAUSE Minutes Remaining: 4.64 at Sample Pt 1 Probe should be at this location

On the 'Data' tab data can be observed in real time. The time between screen update of the averages is set by the data acquisition time parameters on the 'Verification Test Conditions' panels. The product of sample rate and number of samples is typically one or two seconds.

2.5.3 TEST CONDITIONS

File Edit Operate Tools Window Help

M5 Sample Run Terminate Test

Computations Data **Test Conditions** pid controls Misc/error

Test Conditions

% H2O Es	1.000	%-Estimate
Dry%CO2	0.000	%-Estimate
Dry%O2	20.800	%-Estimate
Tstd	20.000	Tstd
Pstd	760.000	Pstd
Port #	1.000	
# of Pts	5.000	
Time-Pt	5.000	Minutes

file path (dialog if empty)
%C:\Program Files\A2000c\data\kwbsamplerun .xls file preferred

PAUSE Minutes Remaining **4.47** at Sample Pt **1** Probe should be at this location

This panel reproduces the stack test conditions that are entered in 'Verify Test Conditions'.

2.5.4 pid controls

M5 Sample Run

Terminate Test

Computations Data Test Conditions **pid controls** Misc/error

PID gains

proportional gain (K_c)	1.000
integral time (T_i , min)	0.000
derivative time (T_d , min)	0.000

dt out (s)

1.125

Isokinetic setpoint

1.00

pidoutput

-1.000

Vout

1.000

Out V Expected

2.500

PAUSE Minutes Remaining: 0.00 at Sample Pt 1 Probe should be at this location

PID GainsProportional Gain (K_c)Integral Time (T_i , min)Derivative Time (T_d , min)**These tuning values are not normally changed.****dt out (s)**

pidoutput

Isokinetic SetPoint**Vout**

Out V Expected

These may be changed during a test (1.00 sets to 100 %).

2.5.5 Misc / error

M5 Sample Run

Terminate Test

Computations Data Test Conditions pid controls **Misc/error**

Avg #'s

Sample Time, msec

Print Time, min

Ven-vac gain

Ven-vac offset

Input Data

Variable		
t-init	2.000	Macc
V-dscfm	0.000	Vm-in
V-meter	0.000	Vol-in
PT-init	1	Pt+Pt0
PAUSE?	☐ Pause	
Data Path	C:\A2000c(2008-10-31)\data\	

Units ?

Vol in	Volume out	Vm in	Vm out
0.62	0.00	0.62	0.00

A2000wf
2010-07-12

error
Pt avg

error
Pt avg 2

error dP@RH

error TEMP

PAUSE Minutes Remaining at Sample Pt Probe should be at this location

These variables are used internally and are displayed elsewhere as needed.

Avg #'s
Sample Time, msec
Print Time, min
VacFactor
QFactor
Error Pt avg
Error Pt avg 2
Error dP@RH
Error TEMP

Units?
Vol in
Volume out
Vm In
Vm out

Input Data
t-init (Macc)
V-dscfm (Vm-in)
V-meter (vol-in)
PT-init (Pt+Pt0)
PAUSE?
Data Path (Data FP)

2.6 View Data

This button allows the operator to view the temperature and transducer readings with the pump and control valve off. No data is logged to the spreadsheet.

M5 Review and Simulation Terminate Review

Computations **Data** Simulation Data Input Test Conditions Misc/error

Analog Data Run Mode Temperature

Live Data

Pitot	55.769	Stack	17.4
Static	0.212	Probe	19.4
Venturi	40.000	Filter Box	19.8
Barometer	753.468	Dryer	20.1
Vacuum	62.720	Filter Out	20.6
NA6	0.000	DGM	21.0
NA7	0.000	aux	1770.0
Ven/Vac	0.000	Console	21.8

Minutes Remaining
1000.53

The 'Simulation Data' panel in 'View Data' can be used to generate artificial data to see the effects of varying temperature, pressure, nozzle diameter, etc. Pitot and vacuum would be set at expected operating conditions. Venturi would be simulated to give the desired sample flow rate. This can be very helpful in determining the optimum nozzle size without inserting the probe or Pitot.

2.6.1 Computation Tab

File Edit Operate Tools Window Help

M5 Review and Simulation Terminate Review

Computations Data Simulation Data Input Test Conditions Misc/error

Computations M

H2O% Vol	0.250	% Est	3.887
Wet Mol Wt	28.805		
Vel, Stack	5.060	M/sec	
Vel, Nozzle	0.000	M/sec	
Isokinetic	0.000	%	
Qv	0.000	DALPM	
Qdslpm	0.000	DSLPM	
Vol (DGM)	0.000	DALiter	
Volume, dsliters	0.000	DSLiter	

Nozzle Diameter 3.887

Pump OFF

Valve Control 3.150

Adjust Nozzle Size for 100 % Isokinetic with Simulation Data

Flow Rate at Venturi
Flow Rate at dry standard conditions
Volume at (optional) DGM conditions
Volume at dry standard conditions

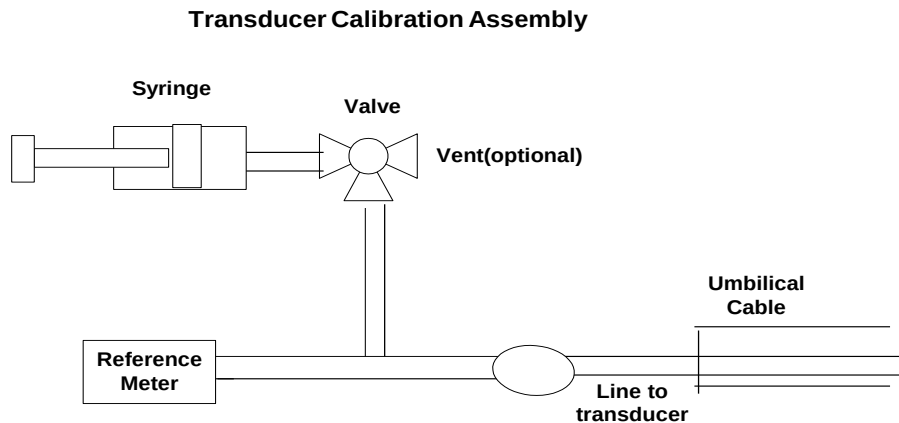
Minutes Remaining
1000.93

On the 'Computations' page, the nozzle diameter can be adjusted for 100% isokinetic operation. This is a convenient way to choose the appropriate nozzle for each test. When in the 'Live Data' mode the Pump button can be turned on and the Valve Control set to the desired flow rate. The calculations may be in English or Metric units based on operator preference.

2.7 Calibrate Sensors

In preparation for transducer calibration the various tubing connections are leak checked and connected to a reference source, such as a water manometer.

A calibration connector kit (A-2000-CPK) is available to help speed the calibrations. The reference meter is not included in the connector kit. It is designed to use the manometer of a manual EPA Method console such as the c-5000, but can be adapted to any reference manometer.



Use the syringe or squeeze bulb to provide calibration pressures. When the valve is shut the Reference Meter and the transducer are maintained at a steady pressure while the data point is obtained.



A-2000-CPK Calibration Connector Kit

2.7.1 Sensor Calibration Routine

Calibration Routine---A2000wf

Instructions

Instructions:
1) Select 'Sensor' and enter the reading of the reference Calibrator for the 'SetPoint'.
***Press 'Clear All Data' button to clear the 'Data' screen.
2) When the 'Computer Voltage' is stable Press 'Collect Data Point' to add it to the regression set.
3) Once satisfied with the regression numbers, hit 'save regression' and these values will be stored to the right.
4) Press 'Clear all data' to delete data set and start a new data set.
5) When finished with all sensors Press 'Save and exit' to save the new calibration values.

(1)

Sensor: Pitot 0

Computer Voltage: 0.000E+0

(2)

Collect Data Point

SetPoint: 0.000

Present Value: -1.219

Percent Error: -Inf

(4)

data regression

(to delete data point, delete the line of text containing that point)

Clear All Data--^

Valve Setting: 0.000 0 to 10 V

Pump: OFF

Slope

Pitot: 25.654

Static: 257.937

Venturi: 51.943

Barometer: 75.006

Vacuum: 12900.76

NA 6: 1.000

NA 7: 1.000

Ven-Vac: 0.000

Stack Temp: 1.000

Probe Temp: 1.000

Filter Box: 1.000

Dryer Temp: 1.000

AuxFilterOut: 1.000

Ext-DGM: 1.000

Aux: 1.000

ConsoleCJT: 1.000

Intercept

Pitot: -1.219

Static: -647.217

Venturi: -5.027

Barometer: 442.544

Vacuum: -254.635

NA 6: 0.000

NA 7: 0.000

NA 8: 0.000

Stack Temp: 0.00

Probe Temp: 0.00

Filter Box: 0.00

Dryer Temp: 0.00

AuxFilterOut: 0.00

Ext-DGM: 0.00

Aux: 0.00

ConsoleCJT: 0.00

(3)

Slope: NaN Intercept: NaN

Save regression-->

Save and exit <F1>

Cancel and exit <esc>

(5)

N Values to average: 20

Time Between Reads(ms): 100

A2000wf-2010-07-10

Before its initial use in a field test, calibrate the Pitot, Venturi and Static pressure transducers using the three-point calibration procedure described in US EPA Method 2F, section 10.3.3. The device shall be recalibrated according to the procedure in US EPA Method 2F, section 10.3.3 no later than 90 days after its first field use following its most recent calibration. At the discretion of the tester, more frequent calibrations (e.g., after a field test) may be performed. No adjustment, other than adjusting to the zero setting, shall be made to the device between calibrations.

Pre-test and post-test calibration checks are performed at amid-range pressure setting. The pressure readings made using the reference device (SetPoint) and the tested device (Present Value) shall agree to within ± 3 percent of the full scale of the tested device or 0.003 inches H₂O, whichever is less restrictive. If this specification is met the test data collected during the field test are valid. If the specification is not met, all test data collected since the last successful calibration or calibration check are invalid and shall be repeated using a pressure-measuring device with a current, valid calibration. Any device that fails the calibration check shall not be used in a field test until a successful recalibration is performed according to the procedures in US EPA Method 2F, section 10.3.3.

TEMPERATURE CONTROLLERS

Temperature controllers shall be calibrated no more than 30 days prior to the start of a field test or series of field tests and recalibrated no more than 30 days after completion of a field test or series of field tests.

2.7.1.1 Pitot Sensor Calibration Routine

1. Select Pitot or Sensor 0

Note:

- The default Pitot sensor reads 0 – 5" H₂O and outputs 0 – 5 Vdc.
 - A minimum of three data points are required to properly calculate the slope and intercept values for each pressure sensor.
2. While the pressure umbilical is disconnected there is no differential pressure so you can collect a SetPoint value of 0.0" H₂O.
 3. Enter 0 in the SetPoint box and click the Collect Data Point button.
 4. Enable the pressure umbilical to deliver a known pressure around 4.0" H₂O through the positive side 3/8" Pitot Quick Connect. Be careful to not exceed its maximum range as this can damage the Pitot sensor. When an inclined manometer is used as the Reference meter, be careful to not exceed its range and blow out the fluid. The negative side of the Pitot has no effect on Pitot calibration and can left open.
 5. Enter the exact known pressure value into the SetPoint box and click the Collect Data Point button.
 6. Enable the pressure umbilical to deliver a known pressure around 1.0" H₂O through the positive side 3/8" Pitot Quick Connect.
 7. Enter the exact known pressure value into the SetPoint box and click the Collect Data Point button.
 8. After a minimum of three data points you may save the data or continue to enter additional known pressures.
 9. If complete Click the Save regression-> button.

Note: This will transfer the Slope and Intercept values of the Pitot Sensor to the table on the right side of the screen.

10. The Pitot Sensor calibration is now complete.

2.7.1.2 Static Sensor Calibration Routine

1. Select Static or Sensor 1

Note:

- The default Static sensor reads +/- 25" H₂O and ideally outputs 0 – 2.50 Vdc when sensing negative pressure and 2.51 – 5.0 Vdc when sensing positive pressure.
 - A minimum of three data points are required to properly calculate the slope and intercept values for each pressure sensor.
2. Click on the Clear All data button to clear out all of the previous values from the previous sensor calibration.
 3. While the pressure umbilical is disconnected there is no differential pressure so you can collect a SetPoint value of 0.0" H₂O.
 4. Enter 0 in the SetPoint box and click the Collect Data Point button.
 5. Enable the pressure umbilical to deliver a known pressure less than 20.0" H₂O through the negative side 3/8" Pitot Quick Connect. Be careful to not exceed its

maximum range as this can damage the Static sensor. The Static sensor can be calibrated in a range less than full scale. The positive side of the Pitot has no effect on Static calibration and can left open.

6. Enter the exact known pressure value into the SetPoint box and click the Collect Data Point button.
7. Enable the pressure umbilical to deliver a known pressure above -20.0" H₂O through the negative side 3/8" Pitot Quick Connect.
8. Enter the exact known pressure value into the SetPoint box and click the Collect Data Point button.
9. After a minimum of three data points you may save the data or continue to enter additional known pressures.
10. If complete Click the Save regression-> button.

Note:

- This will transfer the Slope and Intercept values of the Static sensor to the table on the right side of the screen.

11. The Static sensor calibration is now complete.

2.7.1.3 Venturi Sensor Calibration Routine

1. Select Venturi or Sensor 2

Note:

- The default Venturi sensor reads 0 – 10" H₂O and ideally outputs 0 – 5.0 Vdc.
- A minimum of three data points are required to properly calculate the slope and intercept values for this differential pressure sensor.

2. Click on the Clear All data button to clear out all of the previous values from the sensor calibration.
3. While the pressure umbilical is disconnected there is no differential pressure so you can collect a SetPoint value of 0.0" H₂O.
4. Enter 0 in the SetPoint box and click the Collect Data Point button.
5. Enable the pressure umbilical to deliver a known pressure around 9.0" H₂O through the positive side of the 1/4" Venturi Quick Connect. Be careful to not exceed its maximum range as this can damage the Venturi sensor. The negative side of the Venturi remains open.
6. Enter the exact known pressure value into the SetPoint box and click the Collect Data Point button.
7. After a minimum of three data points you may save the data or continue to enter additional known pressures.
8. If complete Click the Save regression-> button.

Note:

- This will transfer the Slope and Intercept values of the Venturi sensor to the table on the right side of the screen.

9. The Venturi sensor calibration is now complete (Refer also to the section on vacuum effect).

2.7.1.4 Barometric Sensor Calibration Routine

The barometer shall be calibrated no more than 30 days prior to the start of a field test of series of field tests.

This routine can be used to test the accuracy of the current calibration and also to perform a new calibration. Clicking on the 'Sensor' button indicator gives a drop-down menu to select the sensor to be calibrated. Also clicking the arrows next to the 'Sensor' indicator brings up or down the next sensor.

1. Select Barometer or Sensor 3

Note: The default Barometer has a factory specified range of 600 mBar to 1100 mBar for an output of 0.1 to 5.1 volts. Since it is not easy to get a wide range of values for barometric pressure an abbreviated procedure is used.

2. Click on the Clear All data button to clear out all of the previous values.
3. In the 'data' field type '0.1, 17.72' and 'enter' for English units.
4. Take the present barometric pressure reading from a Reference Barometer.
5. Put this value into the SetPoint box and click the Collect Data Point button.
6. Click the Save regression-> button. This will transfer the Slope and Intercept values of the Barometric sensor to the table on the right side of the screen.

The Barometer sensor calibration is now complete.

1.7.1.5 Vacuum Sensor Calibration Routine

1. Select Vacuum or Sensor 4

Note:

- The default Vacuum sensor reads 0 – 20" PSID and ideally outputs 4 – 20 mA.
 - A minimum of three data points are required to properly calculate the slope and intercept values for this differential pressure sensor.
2. Click on the Clear All data button to clear out all of the previous values from the sensor calibration.
 3. While the pressure umbilical is disconnected there is no differential pressure so you can collect a SetPoint value of 0.0" H₂O.
 4. Enter 0 in the SetPoint box and click the Collect Data Point button.
 5. Enable the pressure umbilical to deliver a known pressure around 9.0" H₂O through the positive side of the 1/4" Vacuum Quick Connect. Be careful to not exceed its maximum range as this can damage the Venturi sensor. The negative side of the Venturi remains open.
 6. Enter the exact known pressure value into the SetPoint box and click the Collect Data Point button.
 7. After a minimum of three data points you may save the data or continue to enter additional known pressures.
 8. If complete Click the Save regression-> button.

Note:

- This will transfer the Slope and Intercept values of the Vacuum sensor to the table on the right side of the screen.
9. The Vacuum sensor calibration is now complete.
 10. When finished with all sensors Press 'Save and exit <F1>' to save the new calibration values to C:\Program Files\A2000wf\config\ Current.cfg. You may also 'Save and exit <F1>' after each sensor calibration.

2.8 Flow Check Calibration and Calibration Procedure

– Critical Orifice

A critical orifice is the preferred device for calibrating the AT250. The theoretical equations used to calculate the Venturi flow rate are discussed in Section 4.4. The critical orifice should be placed between the umbilical adapter and the vacuum pump as shown below. The critical orifice has been calibrated previously, to determine K', and the flow rate is given by

$$Q_{cr} = K' * P_{cr} / \sqrt{T_{cr}}$$

In the setup below, the last impinger is separated from the Umbilical Adapter, and P_{cr} is given by $P_{cr} = P_{bar} - dP_{ven}$. Vacuum gauge should be about 18 inHg (1-2 inHg above critical). In 'View' you can observe and compare Q_{cr} with AT250(Q_{dscfm}). Also a detailed calibration can be performed by running the 'Test Program' to get the data below. However the day to day calibrations are quickly performed using the self-guided program accessed by the Menu button 'Critical Orifice Flow Calibration'.

	Barometric	Venturi	Temp	Ref	Ref	Ref	AT250		
	Pressure	dPven	Tcr	Qcr	Time	Vcr	Vol(DSCF)	Diff	Diff
	inHg	inHg	deg R	DSCFM	Min	DSCF	DSCF	DSCF	Percent
Run 1									
Run 2									
Run 3									
Average									

The pretest calibration check requires three runs that bracket the expected flow rates during the test. Each run must agree within + or – 2 percent or the Venturi must be recalibrated. For recalibration use the average of the three or more runs to calculate a new k value.

$$k_{new} = k_{old} * (V_{cr}) / (AT250 \text{ Vol}(DSCF))$$



Critical Orifice with Vacuum Gauge

Included with **A-2000-COK Orifice calibration Kit**

2.8.1 Set Up Using Critical Orifice with Vacuum Gauge

The posttest calibration check requires three runs near the average flow rate of the test. If these runs do not agree within + or – 5 percent or the test is either voided or the lowest volume calculations must be used. In some cases you may be allowed to rerun the posttest calibration check.



Critical Orifice Calibration Setup

2.8.2 Critical Orifice Flow Calibration Button on the Menu

Clicking on this button takes the user through a quick self-guided calibration as shown in the following screen views.

Critical Orifice Flow Calibration_ Manual Flow Valve Control

pid control

BESC 2011-09-14

PID gains

proportional gain (Kc) 0.800

integral time (Ti, min) 0.000

derivative time (Td, min) 0.000

Units ?

English

Data Path

C:\BESC(2011-09-14)\data\
_OrificeFlowCal\VenCalCO-19(2011-
09-15).xls

Place the Critical Orifice assembly between the Umbilical Adapter and the Vacuum Pump line.

Disconnect the last impinger from the Umbilical Adapter.

Press OK when your setup is complete

OK Terminate

Critical Orifice Flow Calibration_ Manual Flow Valve Control

pid control

BESC 2011-09-14

PID gains

proportional gain (Kc) 0.800

integral time (Ti, min) 0.000

derivative time (Td, min) 0.000

Units ?

English

Data Path

C:\BESC(2011-09-14)\data\
_OrificeFlowCal\VenCalCO-19(2011-
09-15).xls

Enter the Orifice coefficient K'.

Enter Data Averaging Time.

Press OK when ready to begin.

Orifice S/N	K'	Averaging Time (MM:SS)
CO-19porch	0.5474	01:00

OK Terminate

Critical Orifice Flow Calibration_ Manual Flow Valve Control

pid control

BESC 2011-09-14

PID gains

proportional gain (Kc) 0.800

integral time (Ti, min) 0.000

derivative time (Td, min) 0.000

Units ?

English

Data Path

C:\BESC(2011-09-14)\data_OrificeFlowCal\VenCalCO-19(2011-09-15).xls

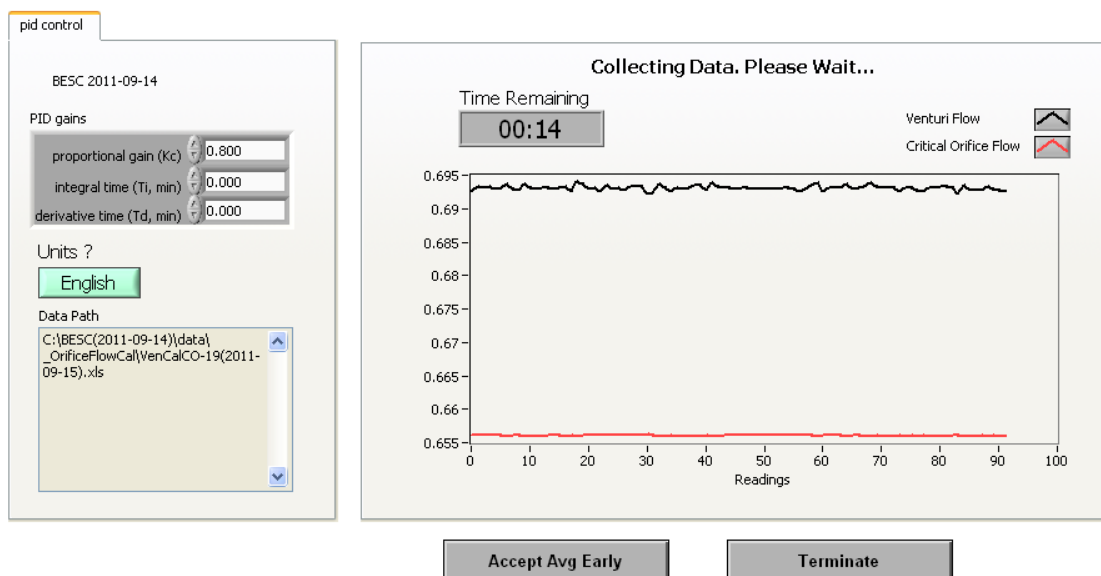
Change Manual Flow Valve % until the vacuum reading of the Critical Orifice assembly is between 17 and 19 inHg.
Input Actual Pressure Reading and Press OK when critical vacuum is reached.

Manual Flow Valve % (0-100)	Flow Rate Actual (DSCFM)	Orifice Flow Rate (DSCFM)	Barometer (inHg)	Venturi Pressure (inH2O)	Venturi Temperature (oF)
58	0.693	0.656	29.445	1.608	79.697

Enter Actual Orifice Vacuum (inHg)

18.000

OK Terminate

Critical Orifice Flow Calibration_ Manual Flow Valve Control

Critical Orifice Flow Calibration_ Manual Flow Valve Control

pid control

BESC 2011-09-14

PID gains

proportional gain (Kc) 0.800

integral time (Ti, min) 0.000

derivative time (Td, min) 0.000

Units ?

English

Data Path

C:\BESC(2011-09-14)\data\
_OrificeFlowCal\VenCalCO-19(2011-
09-15).xls

Old K 0.711

New K 0.673

% Diff 3.802



Warning: New K is more than 2% from Old K.

Would you still like to use NEW K?

Yes No

In most cases a single Orifice can perform the calibration check in a couple of minutes. The spreadsheet data output is shown in the table below. It can be found in the data folder of the program folder with the filename that was assigned previously.

Venturi Calibration Sheet - Critical Orifice

Filename:	knCO24-2011-12-02.xls		
12/2/2011	3:43 PM	Critical Orifice Flow Calibration	
Variable Name	Units	Value	
Critical Orifice Serial Number	-	UAV1622	
Critical Orifice K'	-	0.6553	
Critical Orifice Temperature	oF	66.2	
Critical Orifice Vacuum	inHg	18	
Critical Orifice Flow Rate	DSCFM	0.852	
Barometric Pressure	inHg	29.91	
Venturi Pressure Drop	inH2O	2.376	
A2000 Flow Rate	DSCFM	0.845	
Average Time	MM:SS	0:30	
Old K	-	0.7	
New K	-	0.7055	
% Difference	%	-0.79	
Saved Changes?	-	No	

When multiple orifice tests are run the Orifice is easily changed in the “Critical Orifice with Vacuum Gauge” accessory. For multiple tests, choose ‘No’ when asked to change k’, then use the average value of the ‘new k’ values. The new value of k is then entered using the ‘Verify Test Conditions’ button.

Alternatively, both k and n may be changed using the optimum k and n procedure described below.

2.8.3 Using Critical Orifice to Determine Optimum k and n

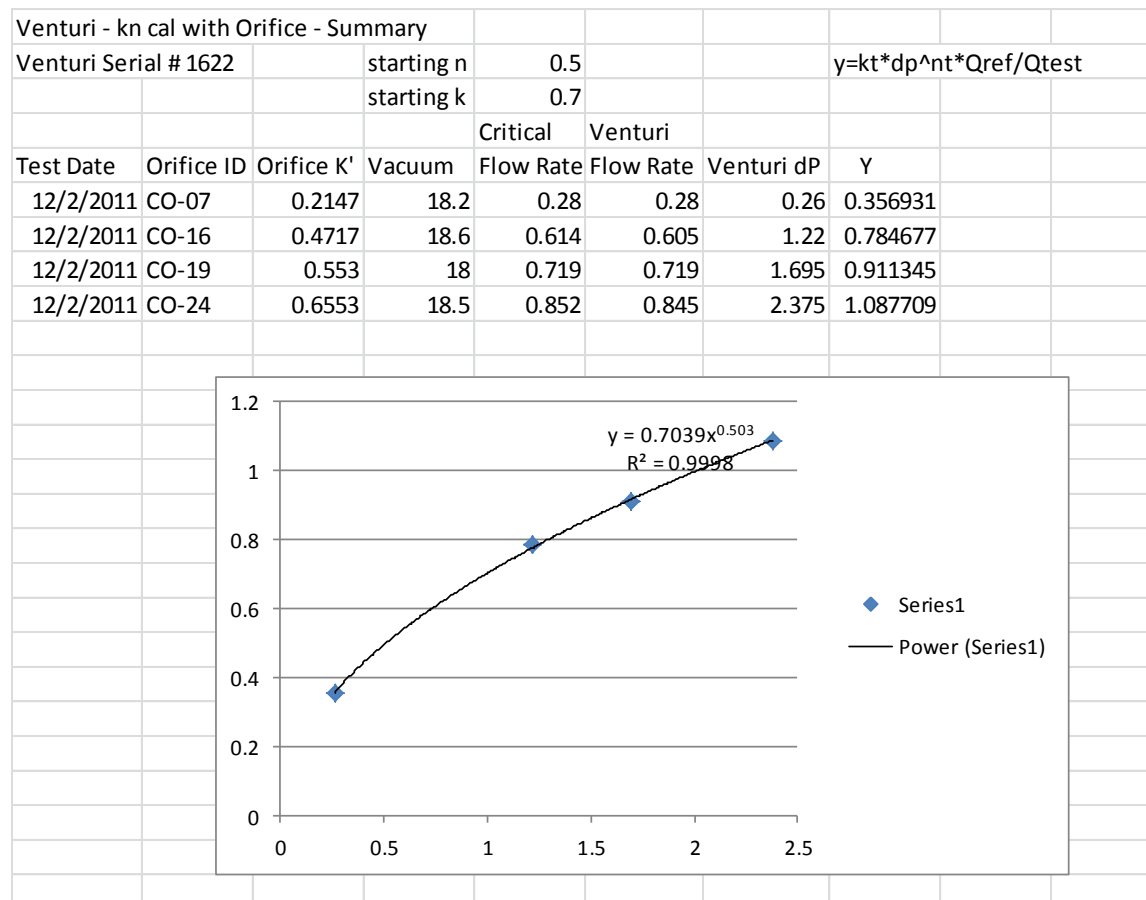
Critical Orifices can be used to calculate empirical values for the Venturi constants k and n. The values of k and n in the ‘Verify Test Conditions’ panel are k1 and n1. The ideal values are k2 and n2. These values are assumed to be related by

$$Q_2 / Q_{\text{test}} = (k_2 * dP^{n_2}) / (k_1 * dP^{n_1})$$
$$\text{Or } Y = (k_2 * dP^{n_2}) = Q_2 / Q_{\text{test}} * (k_1 * dP^{n_1})$$

Q_{test} is the value of A2000(or AT250) Flow Rate. Q₂ is the value of Critical Orifice Flow Rate. And dP is the recorded Venturi differential pressure. A plot of Y vs dP gives a curve with the constants k₂ and n₂.

This calculation is not done in the program but may be set up in an Excel Spreadsheet or equivalent. An example is shown below and may be available in a CD.

Spread sheet and Graph for Optimum k and n values



The above result gives the optimum values of $k = 0.7039$ and $n = 0.503$ over the range of the four critical orifices.

2.9 Flow Check Calibration and Calibration Procedure - Dry Gas Meter

The theoretical equations used to calculate the Venturi flow rate are discussed in Section 4.4. These values are converted to volume at DGM conditions using the ideal gas relation for temperature and pressure. The program posts these values as 'Volume(DGM)' for comparison to the reference meter values.

2.9.1 Pre-test and Post-test Calibration Set Up with Dry Gas Meter Reference

Connect the outlet of the reference DGM in front of the Venturi and Umbilical Adapter with no flow restriction on the inlet side of the DGM. With this set-up as shown below, the readings of the reference DGM can be compared to those of the AT250 to determine if a change has occurred in the value for **k** of the Venturi.

(Alternatively the DGM could be placed on the exhaust side of the control console with the orifice or any other impediment removed, otherwise you would have to correct your readings for the pressure drop).

The initial calibration check requires three runs that bracket the expected flow rates during the test. Each run must agree within + or – 2 percent or the new **k** value for the Venturi must be used. For recalibration use the average of the three runs to calculate a new **k** value. A 'detailed calibration can be performed by running the 'Test Program' to accumulate at least 5 CuFt and get the data below. However the day to day calibrations are quickly performed using the self-guided program accessed by the Menu button 'Dry Gas Meter Flow Calibration'

	Reference	A-2000		
	DGM Reading	Volume(DGM)	Difference	Difference
	ACF	ACF	ACF	Percent
Run 1				
Run 2				
Run 3				
Average				

$$k_{\text{new}} = k_{\text{old}} * (\text{DGM Reading}) / (\text{Volume (DGM)})$$

The post-test calibration check requires three runs near the average flow rate of the test. If these runs do not agree within + or – 5 percent or the test is either voided or the lowest volume calculations must be used. In some cases you may be allowed to rerun the post-test calibration check.



Dry Gas Meter Calibration Setup

2.9.2 Dry Gas Meter Flow Calibration Button on the Menu

Clicking on this button takes the user through a quick self-guided calibration as shown in the following screen views.

Venturi Calibration with Dry Gas Meter

Place Reference Gas Meter at inlet to Venturi Umbilical Adaptor
as shown in User Guide.

Set Reference Meter Serial Number and Number of Calibration
Points

Press OK when your setup is complete

Reference Meter
S/N

534840

Number of
Calibration
Pts

3

OK

Terminate

Venturi Calibration with Dry Gas Meter

Enter the Volume to be accumulated at this Calibration Point

Enter the Flow Rate for this Calibration Point.

Enter the Initial Reference Meter(or DGM) Reading.

Press OK when ready to begin.

Accumulation
Volume
(cubic feet)

10.000

Flow Rate
Set Point
(DSCFM)

0.750

Initial DGM
(cubic feet)

322.372

Pt

1

of

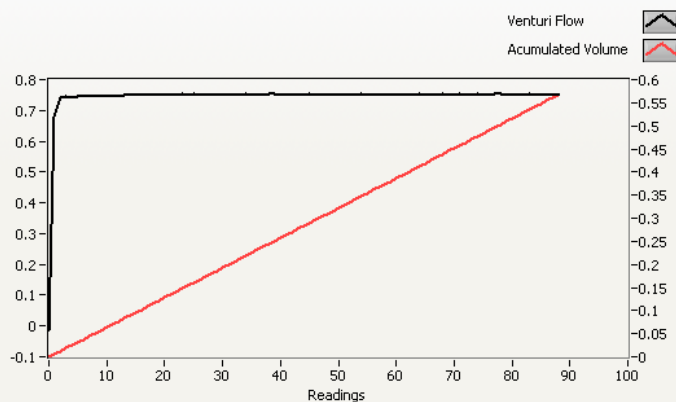
3

OK

Terminate

Venturi Calibration with Dry Gas Meter

Collecting Data. Please Wait...



Flow Rate
(DSCFM)

0.751

Barometer
(inHg)

29.466

Venturi
Pressure
(inH2O)

1.944

Venturi
Temperature
(oF)

79.080

Accept Avg Early

Terminate

Venturi Calibration with Dry Gas Meter

Enter Final Reference Meter(DGM) Reading

Final DGM
(cubic feet)

332.334

OK Terminate

Venturi Flow Calibration with Reference Gas Meter

BESC 2011-09-14

Collection Point	Flow Rate Set Point	Init DGM	Final DGM	DGM Volume	Calculation Volume	Volume % Difference	old k	new k
1	0.5	756.323	766.306	9.983	10.0005	0.17526	0.70441	0.70318
2	0.9	766.306	776.331	10.025	10.005	-0.19971	0.70441	0.70582
3	0.7	776.331	786.343	10.012	10.0017	-0.10286	0.70441	0.70513
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

New K within 2% of Old K.
Would you like to use NEW K?

Old K
0.704

New Avg K
0.705

% Diff
-0.043

Yes No

Spreadsheet Data Output

Filename:	compareDGM(2011-10-19).xls							
10/19/2011	10:39 AM	Reference Gas Meter Flow Calibration						
Variable Name	Units	Value						
Reference Meter S/ N	-	16300942						
Barometric Pressure	inHg	29						
Venturi Pressure Drop	inH2O	1.685						
Old K	-	0.704						
New K	-	0.705						
% Difference	%	-0.04						
Saved Changes?	-	No						
Calibration Point Data	Flow rate	DGM-Start	DGM-Final	Reference Vol	A-2000(DGM)	Diff	Old k	New k
Run #	CFM	CuFt	CuFt	CuFt	CuFt	%	**	**
1	0.5	756.323	766.306	9.983	10.0005	0.1753	0.7044	0.7032
2	0.9	766.306	776.331	10.025	10.005	-0.1997	0.7044	0.7058
3	0.7	776.331	786.343	10.012	10.0017	-0.1029	0.7044	0.7051

For multipoint calibrations, the average of all test runs is used to calculate the new k value.

3 MAINTENANCE

3.1 Control Console

Periodic inspection of the A-2000 Auto-Isokinetic Flow Integrator components is essential to maintain proper operation of the system. Regular maintenance will insure a long system lifetime.

3.2 Maintenance Access

The A-2000 Auto-Isokinetic Flow Integrator components are designed to be easily accessible by removal of the rear cover for periodic inspections and maintenance. The front and rear covers can be removed by releasing the two cover latches. It is seldom necessary to remove the casing wrap. However, if this becomes necessary, first unscrew the component panel screws located on the sides and top of the casing wrap. Then, unscrew the six base securing screws and remove the casing wrap by lifting straight up with care not to disturb the remaining assembly. To maintain stability the screws securing the console panel to the base should remain in place.



Check the general condition of all components and make necessary repairs.

3.2.2 Quick Connects

Keep quick connects clean by wiping with a clean rag and lubricating with a drop of penetrating oil.

3.2.3 Pitot Tube Lines

Check the Pitot tube line quick connects for obstructions by blowing gently, the Pitot tube gauge should respond.

3.2.4 Temperature Controllers

Your system will have two programmable digital controllers. Each temperature controller employs an indicator light to show when the controller is powering its 25 Amp solid-state relay, to indicate heating to the desired SetPoint. When the SetPoint is reached, the controller will cycle off and repeat the on-off cycle in response to thermocouple feedback.

Setting or resetting the temperature for the programmable, digital controller is outlined below. For more details on the operation of the digital controller please refer to the manual included with your console.

3.2.4.1 RESETTING THE TEMPERATURE CONTROLLER

1. You will see four buttons on the front of the controller; from left to right:  **INDEX**,  **UP ARROW**,  **DOWN ARROW**, and  **ENTER**.

2. First hold down the  **ENTER** and the  **UP ARROW** keys. Pressing these keys simultaneously brings up the secondary menu starting at the alarm, tune, or cycle item (depending on programming)

While in the Primary or Secondary Menu, if no key is pressed for a period of 30 seconds, the display will return to the **HOME** position displaying the temperature value.

3. **EDIT** and a **0** will appear, use the  **UP ARROW** key and change the **0** to a **3** then press the  **ENTER** key.

Setting Edit for 3 allows access to both the Secondary and Secure Menus for both Zone 1 and Zone 2. Each menu presents the menu items for Zone 1 first, followed by the menu items for Zone 2, followed by any common menu items that may be present.

4. Enter the SECURE MENU by holding the  **ENTER** and  **UP ARROW** keys for 5 seconds. **SEC1** and a **4** will appear
5. Use the  **INDEX** key to cycle to the next menu that will set the Zone 1 input type to Type K Thermocouple.
6. **InP1** with a **J-1C** above it will appear. Press the **UP ARROW** key to change the **J-1C** to **CA**. Press the  **ENTER** key.
7. Use the  **INDEX** key to cycle through to the SetPoint Low menu. The SetPoint Low menu displays **SPL1** with a **-100** above it. Use the  **UP ARROW** key to change the **-100** to **32** and press the  **ENTER** key.
8. Use the  **INDEX** key to cycle through to the SetPoint High menu. The SetPoint High menu displays **SPH1** with a **1600** above it. Use the  **DOWN ARROW** key to change **1600** to **450** and press the  **ENTER** key.

9. Use the  **INDEX** key to cycle through the menus until you find one that displays **COPY** with **nonE** above it. Use the  **UP ARROW** key to change **nonE** to **1to2** and press the  **ENTER** key.
10. Press the  **INDEX** key one time. The display should now be **OFL** on the top and bottom and top may flash between **2509** and **OFL**.

3.2.5 Manual Leak Check

The automated program leak test procedure described in the previous section is usually adequate, but the system can be manually leak checked as well. The following description may be helpful in understanding the process.

3.2.5.1 Manual Leak-Check with A-2000 Console

With the nozzle plugged, pull a vacuum on the system at least 2 inHg greater than the expected operating vacuum during the test. Any leak indicated must be less than 0.02 cfm.

3.2.5.2 Optional Manual Method

Close the valve on the Venturi-umbilical Adapter. Turn pump on. With the nozzle plugged, open the valve slightly until vacuum gauge reads approximately 2" greater than the expected operating vacuum, and then close the valve. Record the vacuum reading. After 1 minute record the vacuum again. A passing leak rate is given by:

$$Leak\ rate = \frac{P2 - P1}{29.92} * V_{sys} \quad cfm < 0.02\ cfm$$

Where V_{sys} is the approximate volume of the closed system including the nozzle, liner, filter, connecting tubes, impingers, and umbilical adapter.

*****Important*** Slowly release the vacuum by unplugging the nozzle.**

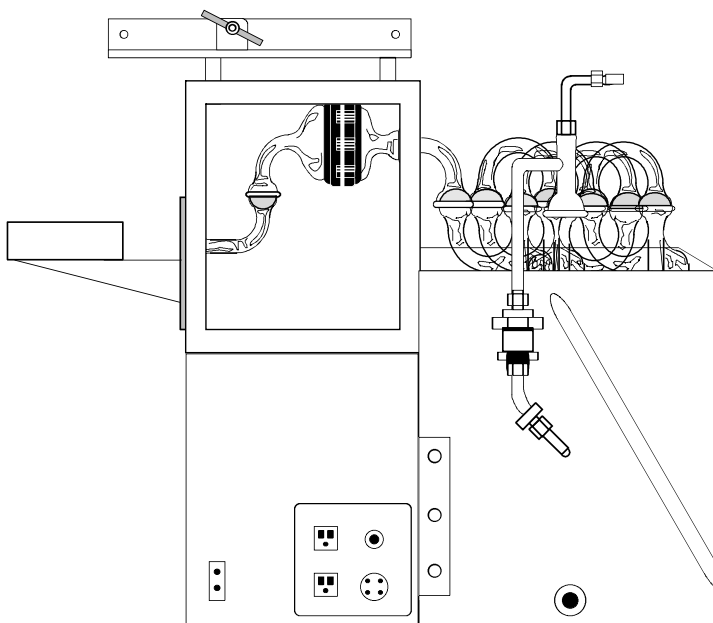
3.3 Sample Case

3.3.1 Heated Filter Compartment

Connect the umbilical between the console and sample case and plug the console into a 115 VAC power source.

Turn on the Filter Heater Control on the A-2000 Auto-Isokinetic Flow Integrator electrical panel. Check its operation by watching the temperature on the controller as it heats. The usual setting is 250°F.

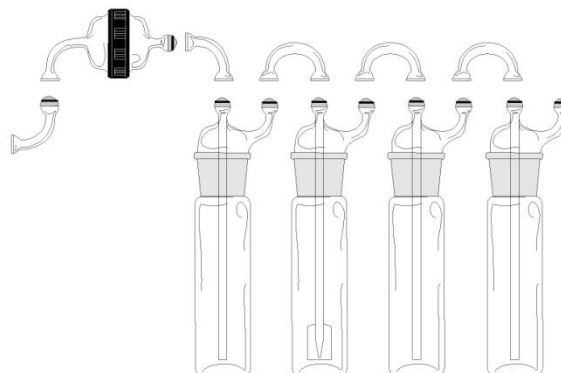
Check the Probe Heater Control in the same manner.



For recording purposes a sample case checklist should be employed to record results of component checking procedures. Each sample case should be numbered. At least two sample cases, including the glassware for each, should be on hand for a stack test. The availability of extra sets increases efficiency during the test since the sample train can be cleaned while the second run is being conducted.

3.3.2 Impinger (Cold) Box

Fill the impinger box with water and check for leaks. Leaks may be repaired with fiberglass and epoxy resin or filled with silicone rubber.



3.3.3 Impinger Glassware

The illustration at left shows the relative position of all glassware during a particulate sample test. It is essential that glassware be clean to insure accurate sampling results.

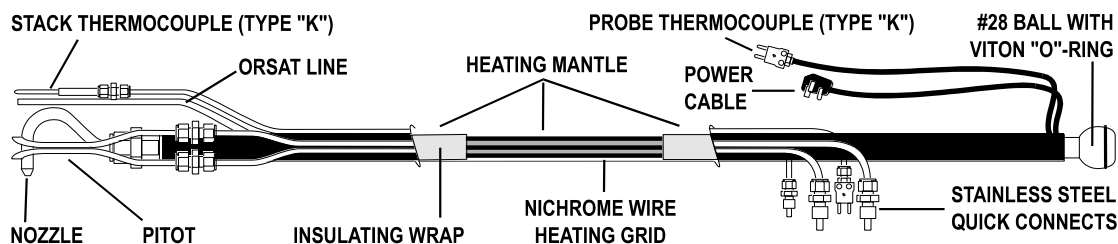
Glass cleaning solution may be used for light cleaning and dichromate sulfuric acid solution for heavy cleaning. Avoid contact of acid cleaning solution with sealing rings or plastic screw caps. Glassware should be rinsed with distilled, de-ionized water and then acetone and allowed to air dry with openings covered to avoid contamination.

3.3.4 Filter Assembly Pressure Drop Check

Connect the filter holder assembly directly to the umbilical adapter and run the umbilical cable to the console for a pressure-drop check. Turn the pump on and adjust flow on the

orifice gauge to a reading of 5.0 inches Hg. If the vacuum gauge on the console reads higher than two inches of mercury, the frit needs cleaning and rechecking.

3.4 Probe Assembly



The probe nozzle is sealed with a front and rear ferrule and a union nut to the union body that fastens to a permanently welded union nut on the sample entry end of the probe sheath. The sample entry end of the probe liner inserts into this union body and is secured with an inverted rear ferrule and Viton O-ring. The end of the probe liner should be positioned flush with the front end of the welded union nut when inserting in the union body. ***Detach the union assembly from the probe sheath welded nut before attempting to remove the probe liner.*** Finger tightening is sufficient to reattach the union body assembly (with inverted ferrule and Viton O-ring) to the welded nut. ***Do not use a wrench for this procedure.*** However the probe nozzle-to-union body attachment requires two wrenches for proper tightening. We suggest using a single graphite ferrule as an alternative to the Viton O-ring when stack temperatures over 500°F are expected.

3.4.1 The Probe Sheath and Probe Liner

The liner must be removed from the probe sheath to inspect the heating mantle or replace the liner. Remove the nozzle and union from the welded nut and gently slide the liner out of the sheath. To reassemble, slide the probe liner inside the probe sheath and guide the end of the liner through the hole of the welded union nut until it extends flush to the end of the nut.

3.4.2 Probe Test

Examine the electrical components of the probe for any visible damage prior to following test procedures.

1. Plug the probe into the sample case and set the probe heat controller to 250°. The probe should become warm to the touch over its entire length in a few minutes. If the probe does not heat, check the probe heat control for loose connections, the umbilical cord and the sample case.
2. Use an ohmmeter to measure the resistance between leads and also to ground. Deviations from proper value indicate faulty wiring.

3.4.3 Probe Cleaning

Clean the probe liner internally by brushing, first using tap water, and then distilled, de-ionized water, and finally using acetone. Perform a final rinse with acetone and allow to air dry. Inspect visually for cleanliness and repeat the procedure if necessary.

Clean the stainless steel union, nut ferrules, Viton O-ring, and nozzle by scrubbing. Rinse with distilled, de-ionized water, and then with acetone and allow to air dry with all open ends of probes and nozzles covered with serum caps or saran wrap to prevent contamination.

3.5 S-Type Pitot Tube Maintenance

Remove the quick connects from the Pitot tube and wipe them clean. A drop of penetrating oil helps keep them in good working condition.

Use compressed air to blow out the Pitot tube. Both the Pitot tube and sheath should be wiped or washed clean with water and air-dried. Inspect the Pitot tube openings for damage and misalignment, and, if necessary, repair and recalibrate the tube.

3.5.1 Nozzle

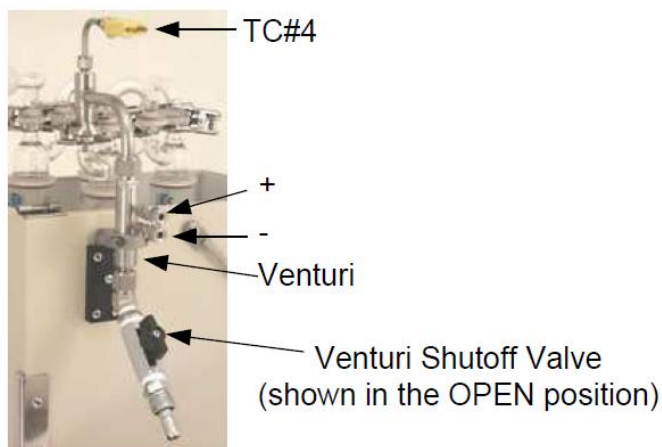
Inspect the knife-edge of the nozzle for damage. If damaged, repair and recalibrate the nozzle (see recalibration section). Clean the nozzle by scrubbing, then rinse with distilled, de-ionized water, and finally with acetone. Allow to air dry with the open ends covered with serum caps to avoid contamination or damage to the knife-edge.

Ideally there should be a selection of nozzles ranging from 1/8" to 1/2" in diameter in 1/16" increments with two or three sets of each size. A number reflecting the exact diameter in 1/32" increments should be etched near the inlet end of each of the nozzles (e.g. #8 indicates a 1/4" nozzle, #9 indicates a 9/32" nozzle).

3.6 Umbilical Cord and Venturi Adapter

The typical umbilical cord consists of one sample (vacuum) line with a 1/2" Quick-connect (QC), two Pitot lines with 3/8" QC's, one Orsat line with a 1/4" QC, one power line with Amphenol connectors, and five thermocouple lines. Lines of like dimension and purpose are numbered at each end to avoid cross connections and assist in proper hookup of the umbilical couplings.

The A-2000 Venturi Adapter fits the same as a typical adapter. It includes two pressure taps and a shutoff valve.



3.6.1 Quick Connects

Wipe the vacuum line and Pitot tube quick connects and apply a drop of penetrating oil on each to keep them in good working condition.

3.6.2 Umbilical Adapter Thermocouple Check

Check the impinger temperature thermocouple ("DRYER" connections and settings on the console) on the umbilical adapter against a mercury-in-glass thermometer.

3.6.3 Sample and Pitot Lines

Test the sample line for leaks by placing a plugged male quick connect into the inlet, connecting the line to the console and checking for leakage as before. If there is leakage, all connections should first be checked, and then the tubing. If the leak is not evident by inspection, slightly pressurize the line and check for leaks using soapy water.

Connect one end of the Pitot tube line to a 36-inch U-tube mercury manometer or a similar device. Pull a vacuum of at least ten inches, seal the tubing at the pump end, and check for leaks by noting the mercury manometer. Do the same for the other side. If there is leakage, check the tubing by following the same procedure as for the vacuum line.

3.6.4 Venturi

Check the Venturi and clean if necessary. Test the pressure lines in the same manner as the Pitot lines.

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4 SYSTEM CALIBRATION & PROCEDURES

4.1 Control Console

The temperature controllers should be checked against a reference Type K thermocouple for accuracy.

Transducers should be calibrated following the requirements of US EPA Method 2F. Linear coefficients can be obtained by multipoint calibration procedures using acceptable reference standards. Calibration checks or full calibrations can be performed quickly in the Sensor Calibration program on the Main Menu.

4.2 Pitot Tube Calibration

Measure the velocity pressure, ΔP , at the same point within a cross section of a straight run of ductwork (8 diameters downstream and 2 diameters upstream from any point of disturbance) with a standard Pitot tube and the S-type Pitot tube for the desired range of gas velocities. The S-type Pitot should be calibrated twice, reversing the direction of the legs during the second calibration. Make at least three determinations for each direction. Determine the Pitot tube coefficient ($C_{p(s)}$) of each direction as:

$$C_{p(s)} = 0.99 \sqrt{\frac{\Delta P_{std}}{\Delta P_s}}$$

Pitot tube acceptability is in the range of $C_p = 0.85 \pm 0.03$ in both directions, since the nomograph used with the sampling train is designed for this number. If the C_p is not in this range, the Pitot tube was probably not calibrated properly. Be sure that the Pitot tube is square with the duct and pointing directly into the gas stream. A special nomograph is available to correct for variations in C_p . (**ES Part Number: M5-SRN**)

4.3 Nozzle Calibration

Using a micrometer or a dial caliper, measure the inside diameter of the nozzle to the nearest 0.001 inch. Make ten separate measurements using different diameters each time and obtain the average of the measurements. The largest deviation from the average should not exceed 0.002 inch. If the variation is more than 0.002 inch, repair the nozzle with a tapered plumb bob (for inside damage) or emery paper (for outside damage). If the nozzle is satisfactory, polish off the old inscribed value, or, if new, etch the value of the inside diameter on the outside of the nozzle.

4.4 Venturi Calibration

Theory Of Operation

The details and procedures for day to day calibrations are shown in Sections 2.8 and 2.9. The following discussion shows how to perform an independent calibration of the Venturi Flow Meter and compare its performance to the theoretical Venturi flow equation. The Venturi-umbilical adapter may be calibrated against a NIST certified Bubble Flow Meter, a Wet Test Meter, a Reference Dry Gas Meter, or a Critical Orifice.

To demonstrate self-consistency, the Venturi may be used to check the calibration of the DGM if the temperature, pressures and molecular weight is known. In these applications the primary source of error is in the differential pressure measurement. This stresses the importance of very accurate pressure measurements. Since the flow rate is proportional to the square root of the differential pressure measurement, the contribution to the flow rate error is $\frac{1}{2}$ the differential pressure error. Errors in the measurement of absolute temperature and pressure also contribute to the flow rate error.

The Venturi coefficients k and n are calibrated using the flow equation

$$Q = k \cdot \sqrt{(T/P/M) \cdot \Delta P^n}$$

A typical spreadsheet and graph for performing this calibration are shown in the figure below.

The theoretical value of n is 0.5. A quick calibration may be done by assigning n a value between 0.5 and 0.52. The value for k is then determined by running a test program with a reference meter in series at the inlet of the system. The value of k is then raised or lowered proportionally to make the Venturi calculated volume equal to the volume registered by the reference gas meter. Past results have given shown accuracy of better than 1 %.

4.4.1 Venturi Calibration Procedure

Most testers can perform a 2-parameter (k, n) calibration by pulling air through the adapter with the typical US EPA Method 5 system and record the data to plot the equations below.

$$Y = Q_a \cdot P_a / T_a \cdot \sqrt{(M \cdot T_v / P_v)}$$

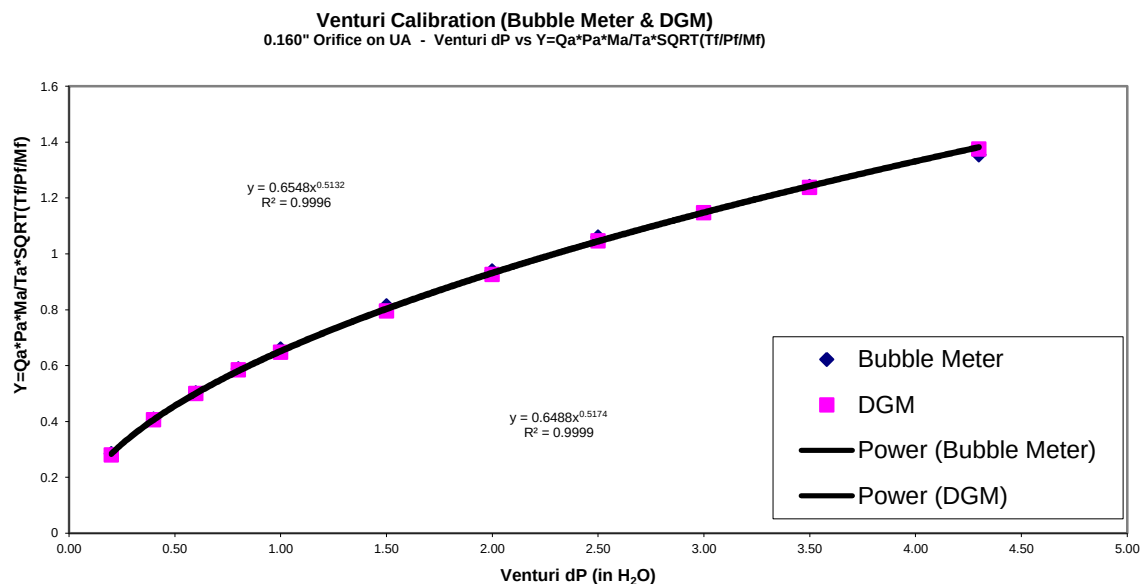
$$X = \Delta P_v$$

Using a spreadsheet plot $Y = k \cdot X^n$ to give the constants k and n.

Where:

- Q_a = flow rate of calibrator, cfm
- P_a = absolute pressure of calibrator, inHg
- T_a = absolute temperature of calibrator, Rankine degrees - °R
- M = Molecular weight of gas
- T_v = absolute temperature of venturi, Rankine degrees - °R

P_v = absolute pressure of venturi, inHg
 ΔP_v = pressure across venturi taps, inH₂O



4.5 Quick Venturi Calibration

Assigning the theoretical value of 0.5 or a value between 0.50 and 0.52 provides good accuracy over the operating range of US EPA Method 5. The value for k can then be determined by running with a reference meter in series at the inlet of the system with its temperature entered into TC Inlet Labeled '6-DGM'. The value of k is then raised or lowered proportionally to make the Venturi calculated volume equal to the volume registered by the reference gas meter.

1. Be sure there are no leaks in the system.
2. Connect the outlet of the Reference Gas Meter to the inlet to the probe or at some convenient point in front of the Sample Venturi located on the Cold Box Umbilical Adapter.
3. Connect a thermocouple from the Reference Gas Meter to the '6-DGM' channel on the A-2000wf Console.
4. Enter the Reference Gas Meter correction factor (Gamma) into the Verify Test Conditions – Cp and Flow Parameters tab.
5. Record the beginning reading if required on the Reference Gas Meter
6. Run a test run long enough to give about 10 cf (280 liters). Less volume can be collected if suitable accuracy is obtained.
7. Record the ending reading if required on the Reference Gas Meter and determine the Reference Volume $Vol_{(ref)}$.
8. Read from the console or look at the data sheet for the test run to obtain $Vol_{(DGM)}$. Be sure that the Dryer temperature and the DGM temperature were recorded properly.
9. If the volume computed by the program varies by more than 2 per cent from the reference meter the value of the Venturi coefficient 'k' should be corrected by the same percent.

10. In the 'Verify Test Conditions' mode find the value of 'k' for the Venturi and designate it 'k(old)'.
11. $k_{(new)} = k_{(old)} * Vol_{(ref)} / Vol_{(DGM)}$.
12. For better accuracy run multiple tests at different flow rates and take the average.
13. In the 'Verify Test Conditions' mode enter k(new) as the value of 'k' for the Venturi.
14. Rerun the test to verify the new calibration. The Venturi should not have to be recalibrated if kept in clean condition.

Daily calibrations and flow checks are best performed using the self-guided programs in section 2.8 and 2.9. Where the program refers to the dry gas meter (DGM), the operator may use other flow references as their situation permits.

4.6 Calculations

4.6.1 Sample Flow Rate and Volume

The sample flow rate is determined by the venturi located after the last impinger and is presumed to be dry. The rate is given by

$$Q_v = k * \Delta P^n * \text{sqrt}(T_v / P_v / M_d)$$

Correct the sample flow rate to standard conditions (70°F, 29.92 inches Hg) by

$$Q_{dscf} = Q_v * T_{std} * P_v / T_v / P_{std}$$

The dry standard volume is obtained by integrating the flow rate over the sampling time.

$$V_{std} = \text{Sum} \{Q_{dscf} * \text{time increment}\}$$

Where:

T_{std}	=	Absolute temperature at standard conditions 530°R.
T_v	=	Absolute temperature at venturi (°F + 460), °R.
P_v	=	Barometric pressure at the venturi meter, in. Hg
ΔP	=	Pressure drop across the venturi meter, in. H ₂ O
P_{std}	=	Absolute pressure at standard conditions, 29.92 inches Hg
M_d	=	dry molecular weight of gas
k and n	=	venturi calibration constants

4.6.2 Volume of Water Vapor

$$V_{Wstd} = V_{1c} \left[\frac{\rho_{H_2O}}{M_{H_2O}} \right] \left[\frac{RT_{std}}{P_{std}} \right] = \left[0.0472 \frac{cu. ft.}{ml} \right] V_{1c}$$

Where:

V_{Wstd}	=	Volume of water vapor in the gas sample (standard conditions), cu. ft.
------------	---	--

V_{1c}	=	Total volume of liquid collected in impingers and silica gel
ρ_{H_2O}	=	Density of water, 1g/ml
M_{H_2O}	=	Molecular weight of water, 18 lb./lb.-mole
R	=	Ideal gas constant, 21.83 in. Hg-cu. ft./lb.-mole-°R
T_{std}	=	Absolute temperature at standard conditions, 530°R
P_{std}	=	Absolute pressure at standard conditions, 29.92 in. Hg

4.6.3 Moisture Content

$$B_{wo} = \frac{V_{Wstd}}{V_{std} + V_{Wstd}}$$

Where:

B_{wo}	=	Proportion by volume of water vapor in the gas stream, dimensionless
V_{Wstd}	=	Volume of water in gas sample (standard conditions), cu. ft.
V_{std}	=	Volume of gas sample (standard conditions), cu. ft.

4.6.4 Stack Gas Velocity

$$(Vs)_{avg} = K_p C_p (\sqrt{\Delta p_{avg}}) \left[\frac{(Ts)_{avg}}{P_s M_s} \right]^{1/2}$$

Where:

$(Vs)_{avg}$	=	Stack gas velocity, feet per second (fps)
K_p	=	85.48 ft. /sec. $\left[\frac{lb.}{lb.mole-°R} \right]^{1/2}$
C_p	=	Pitot tube coefficient, dimensionless
$(Ts)_{avg}$	=	Average absolute stack gas temperature, °R
(ΔP_{avg})	=	Average velocity head of stack gas, inH ₂ O
P_s	=	Absolute stack gas pressure, inHg
M_s	=	Molecular weight of stack gas (wet basis), lb/lb mole
M_d	=	Dry molecular weight of stack gas, (28.8 for air)
B_{wo}	=	Proportion by volume of water vapor in the gas stream

4.6.5 Volumetric Flow Rate

$$Q_s = 60(1 - B_{wo}) V_s A \left[\frac{T_{std}}{(Ts)_{avg}} \right] \left[\frac{P_s}{P_{std}} \right]$$

Where:

Q_s	=	Volumetric flow rate, dry basis, standard conditions, ft ³ /min.
A	=	Cross-sectional area of stack, ft ²
T_{std}	=	Absolute temperature at standard conditions, 530°R
P_{std}	=	Absolute pressure at standard conditions, 29.92 in. Hg

4.6.6 Total Particulate Weight

Determine the total particulate catch from the sum of the weights from the analysis.

Concentration in gr/scf

$$C's = 0.0154 \frac{gr}{mg} \left[\frac{M_n}{V_{std}} \right]$$

Where:

- $C's$ = Concentration of particulate matter in stack gas, gr./scf, dry basis
 M_n = Total amount of particulate matter collected, mg.
 V_{std} = Volume of gas sample through dry gas meter (standard conditions), cu. ft.

4.6.7 Emission Rate in Pounds/Hour

Where:

- G = Stack emission rate, lbs/min.
 $C's$ = Concentration of particulate matter in stack gas, gr/scf, dry basis
 Qs = Volumetric flow rate, dry basis, standard conditions, ft³/min.

4.6.8 Percent Isokinetic

The Isokinetic calculations performed during the stack test use the estimated moisture content. When the actual moisture content is determined in the lab, the true Isokinetic calculation can be done.

$$I_{(true)} = I_{(calc)} * \{1 - Bwo_{(est)}\} / \{1 - Bwo_{(true)}\}$$

When $I_{(calc)}$ is controlled to 100 percent:

$$\text{Percent of isokinetic sampling } I = 100 * \{1 - Bwo_{(est)}\} / \{1 - Bwo_{(true)}\}$$

4.6.9 Acceptable Isokinetic Sampling Results

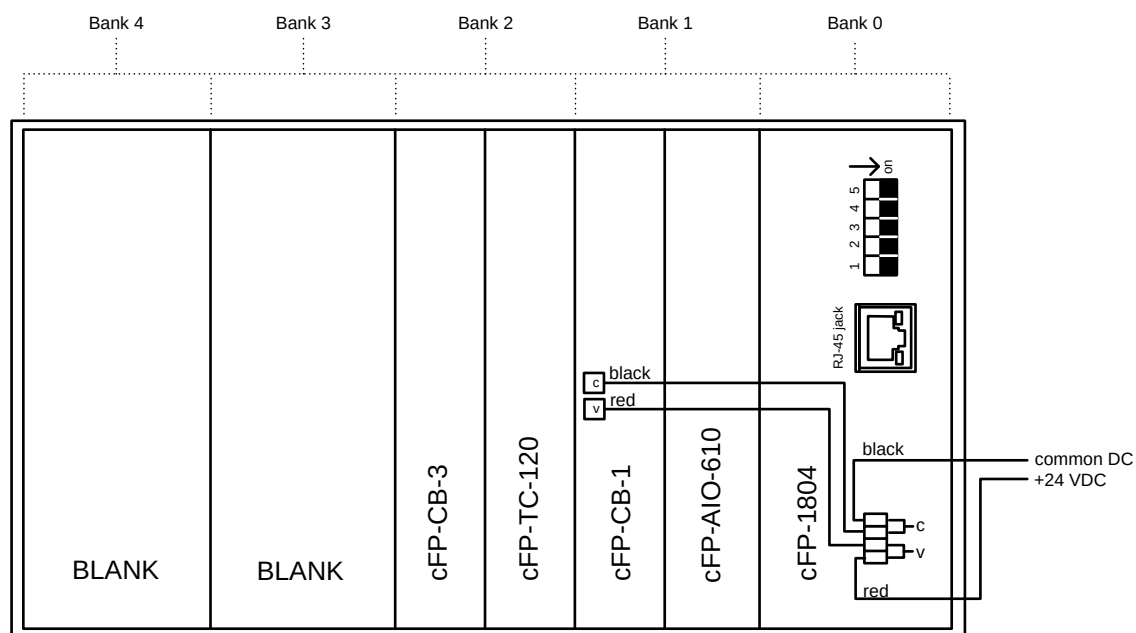
If $90\% \leq I \leq 110\%$, the results are acceptable, otherwise, reject the results and repeat the test.

5

These drawing representations are given to help in setting up and troubleshooting the A-2000 Auto-Isokinetic Flow Integrator system.

5.1 cFP-1804

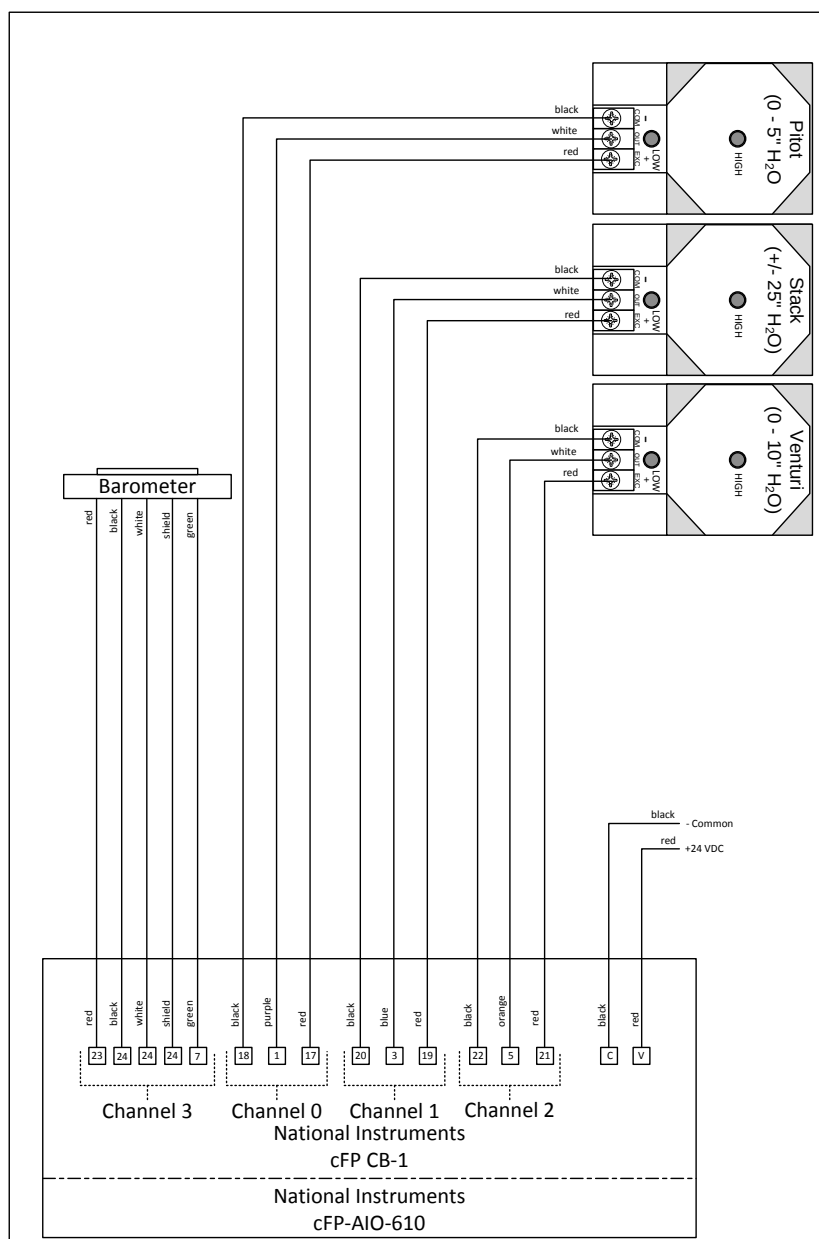
5.1.1 cFP-1804 Module



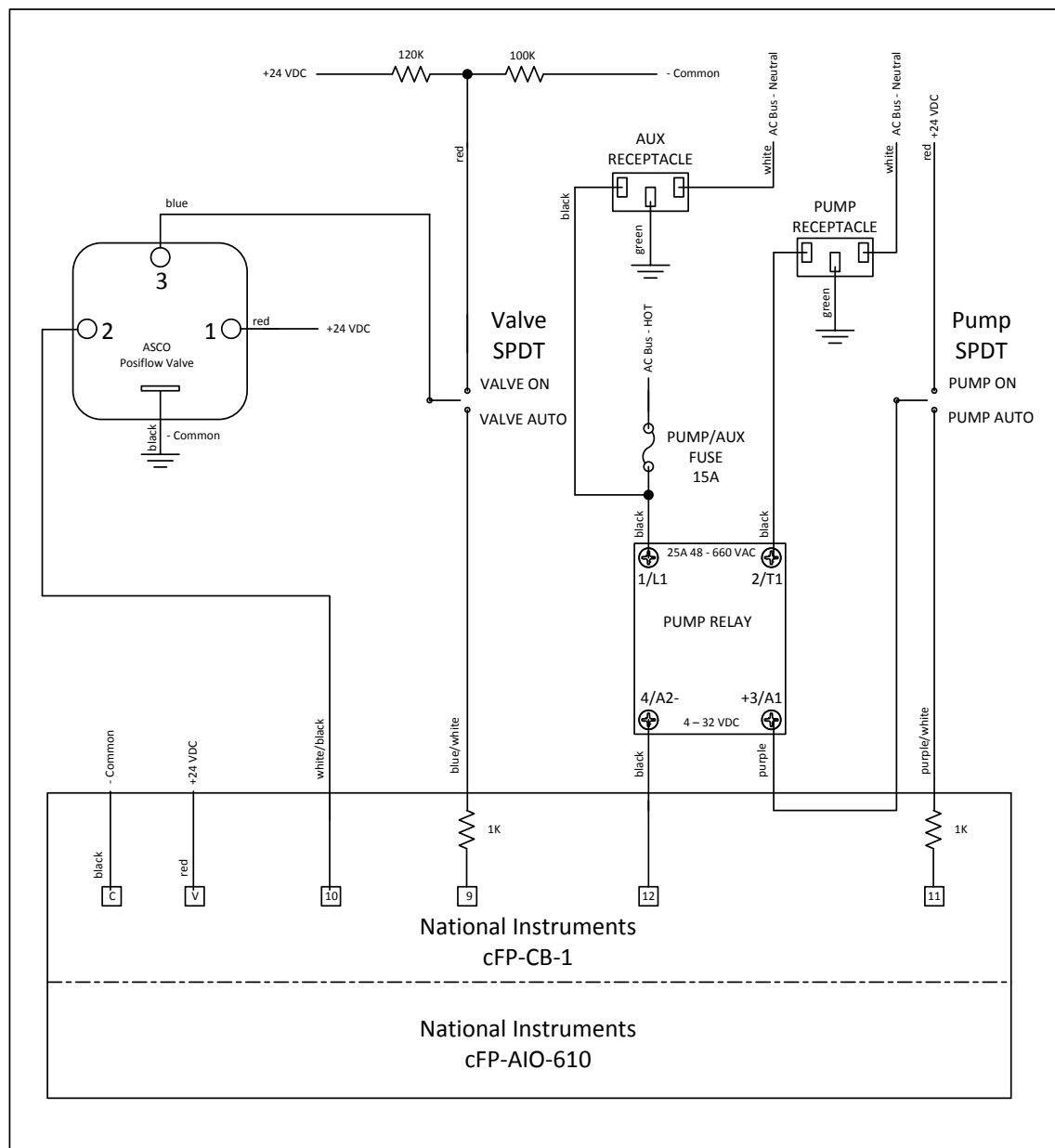
cFP-1804 DIP SWITCH SETTINGS

	IP RESET	CONSOLE OUT	BAUD RATE			BAUD
DIP SWITCH	1	2	3	4	5	
	0 NO IP RESET	0 ETHERNET OUT	0	0	0	300
	0 NO IP RESET	1 RS-232 OUT	0	0	1	1200
	1 IP RESET	0 ETHERNETOUT	0	1	0	2400
	1 IP RESET	1 RS-232 OUT	0	1	1	9600
			1	0	0	19200
			1	0	1	38400
			1	1	0	57600
			1	1	1	115200

5.1.2 cFP-AIO-610 Analog Input Channel Connections

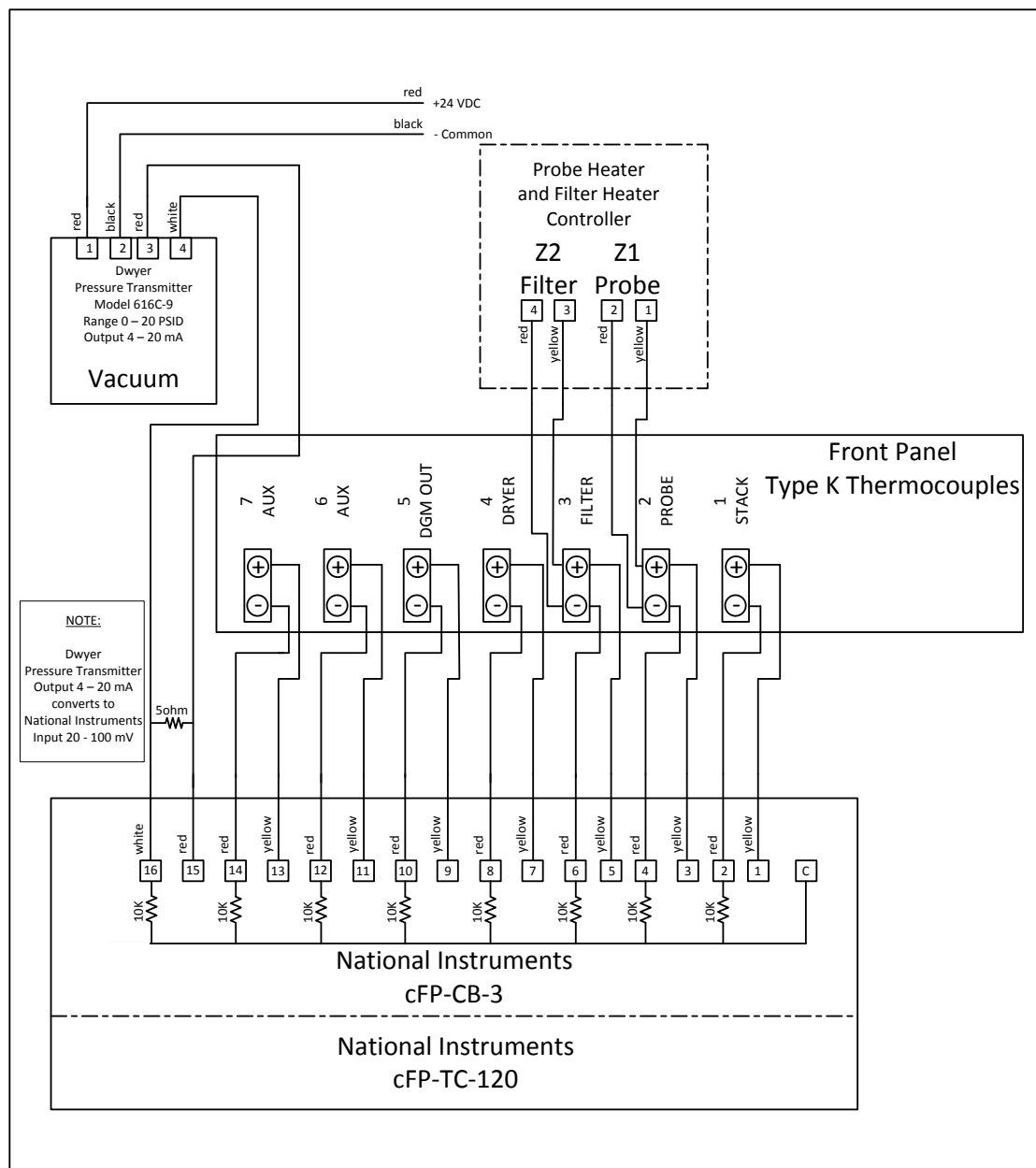


5.1.3 cFP-AIO-610 Analog Output Channel Connections



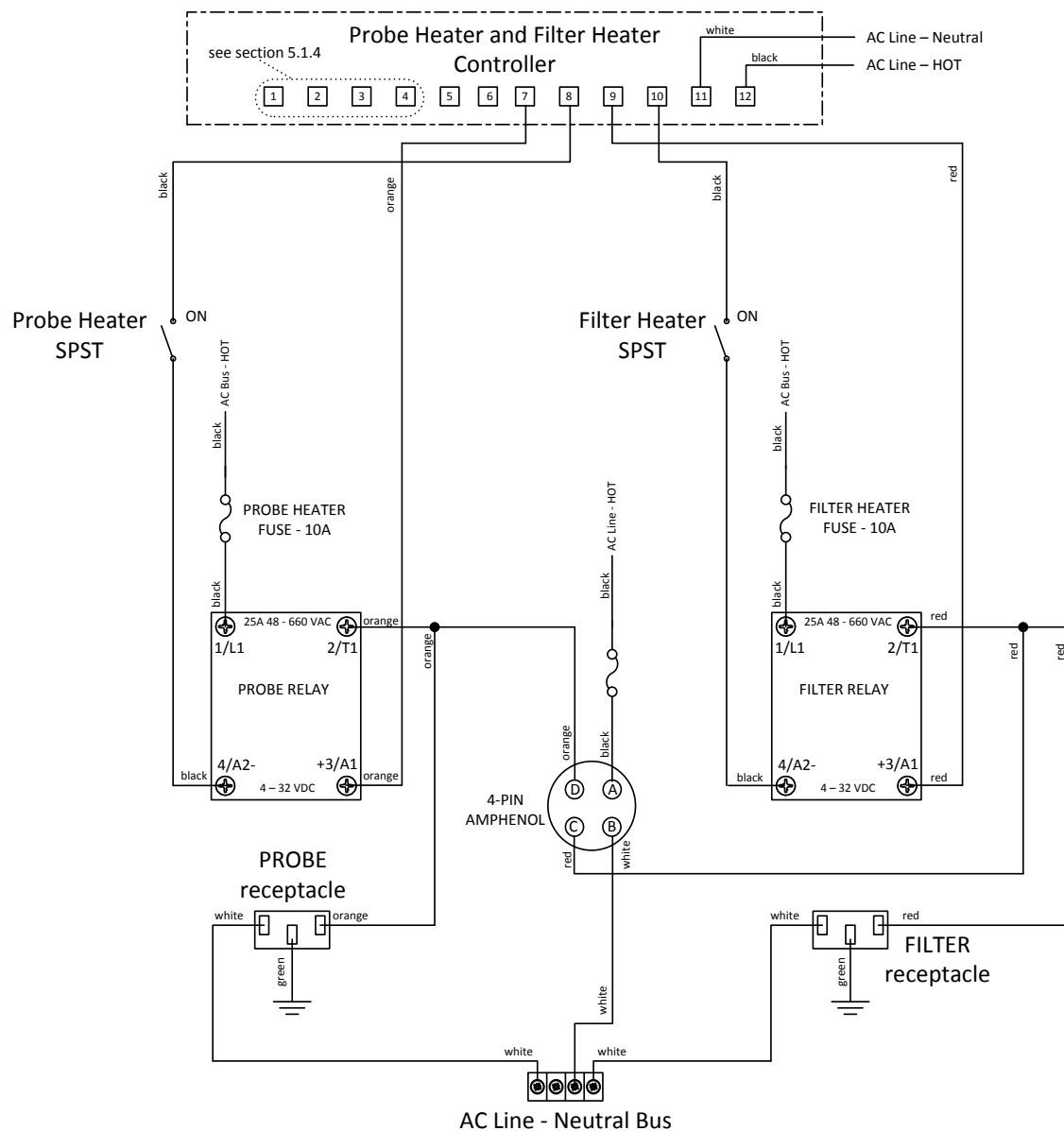
NI cFP-AIO-610 (Analog 4 - IN, 4 - OUT)			
NI Channel	Item	Scale	Range
0 - IN	Pitot	0 - 5" H ₂ O	0 - 5 V
1 - IN	Static	+/- 25" H ₂ O	0 - 5 V
2 - IN	Venturi	0 - 10" H ₂ O	0 - 5 V
3 - IN	Barometer	18 - 30" Hg	0 - 5 V
4 - OUT	Valve	0 - 100%	0 - 10 V
5 - OUT	Pump	0 - 100%	0 - 10 V
6 - OUT	AUX	-	0 - 10 V
7 - OUT	AUX	-	0 - 10 V

5.1.4 cFP-TC-120 Thermocouple Connections



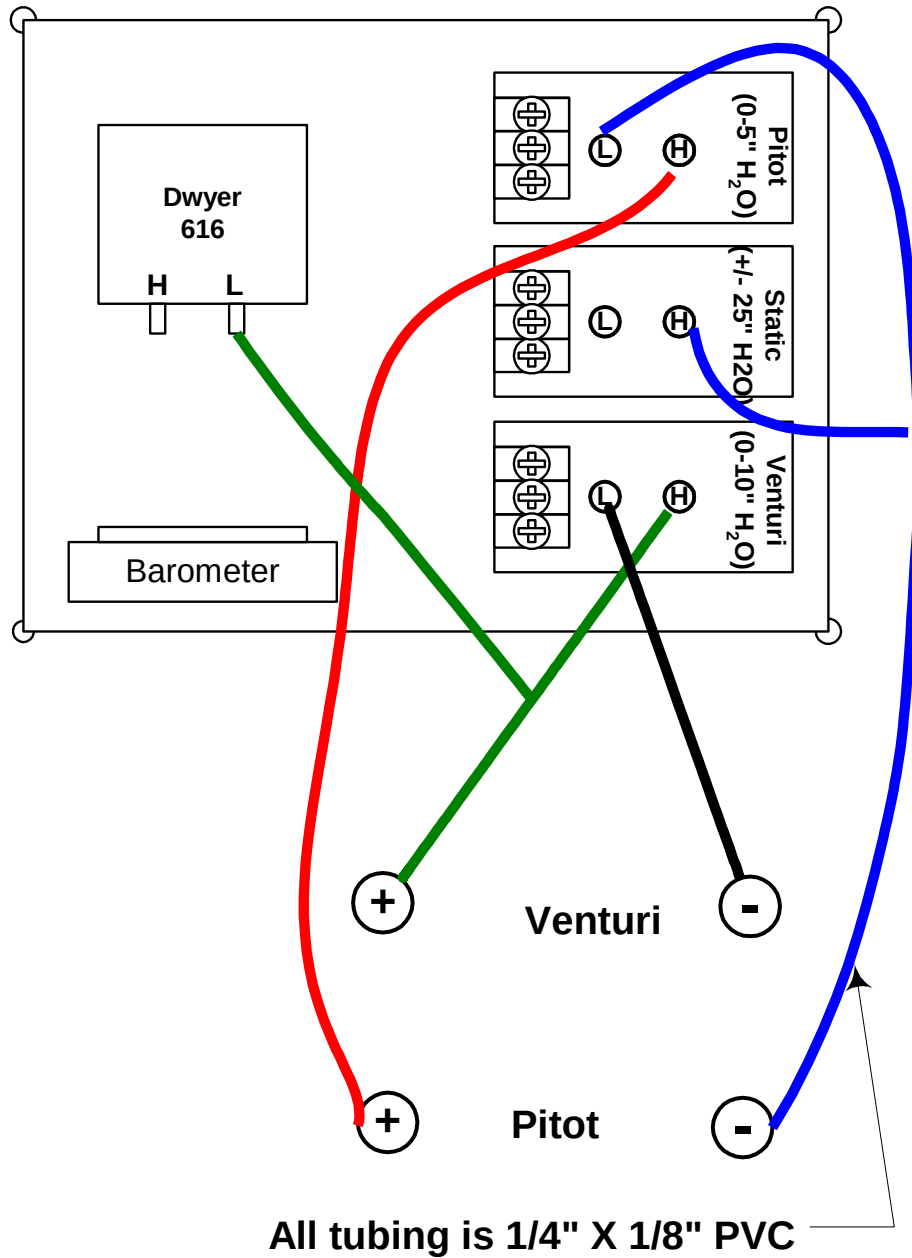
NI cFP-TC-120 (T/C in)				
NI Channel	Item	Terminal		Type
0	1 - STACK	1	+	K
		2	-	
1	2 - PROBE	3	+	K
		4	-	
2	3 - FILTER	5	+	K
		6	-	
3	4 - DRYER	7	+	K
		8	-	
4	5 - AUX	9	+	K
		10	-	
5	6 - EXTERNAL DGM	11	+	K
		12	-	
6	7 - AUX	13	+	K
		14	-	
7	Vacuum (inHg)	15	+	-20 to +80mV
		16	-	
8	"BOX - CJT"			

5.1.5 Heater Wiring



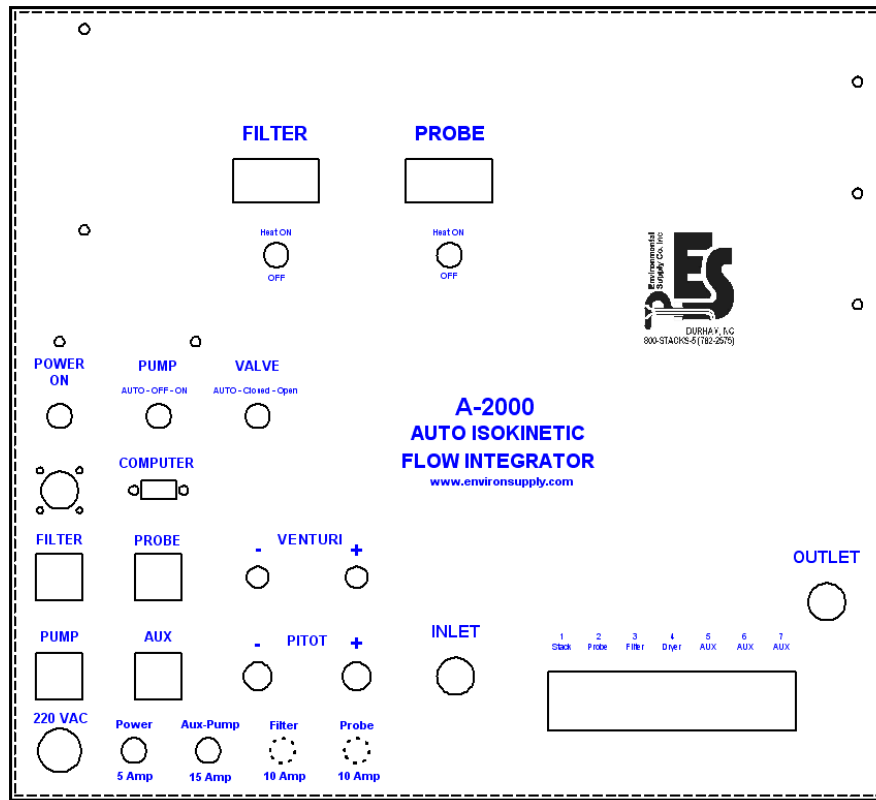
5.4 Transducer Plumbing

Plumbing Rear View



Transducer Plumbing		
Item	Port	Connector
Pitot	High	Pitot +
	Low	Pitot -
Static	High	Pitot -
	Low	OPEN
Venturi	High	Venturi +
	Low	Venturi -
Vacuum	High	OPEN
	Low	Venturi +

5.5 Front Panel Layout



6.1 SOFTWARE INSTALLATION

*****Before installing a new program*****

*****be sure that you have a backup copy of 'config' and 'data' folders*****

*****in order to preserve the hardware calibrations and previous data*****

Insert the installation CD and run the set up program:

The installation is self-guided and automatic and should be ready to run with an Icon on the front panel. 'AT250' may be used in place of 'A2000wf' folder name

The following details are included only for reference purposes.

D:\A2000wf(yyyd-mm-dd)Installer\volume\setup.exe; *where D: is the drive letter of the CD drive.*

Follow the instructions of the setup program and it is recommended that you use the default file paths. National Instruments Measurement and Automation Explorer (MAX) will be installed as well as the Labview Runtime Engine.

The following files and directories are created along with a number of support files:

```
C:\Program Files\A2000wf \A2000wfxxxx.exe  
C:\Program Files\A2000wf \A2000wfxxxx.ini  
C:\Program Files\A2000wf \A2000wfxxxx.aliases  
C:\Program Files\A2000wf \data  
C:\Program Files\A2000wf \config\current.cfg  
C:\Program Files\A2000wf \config\initial.cfg  
C:\Program Files\A2000wf \config\A2000wf.iak  
C:\Program Files\A2000wf \Support\FieldPoint.iak
```

A re-installer without National Instruments Measurement and Automation Explorer (MAX) and the Labview Runtime Engine can also be loaded in a few minutes by inserting the installation CD and running the set up program:

D:\A2000wf(yyyy-mm-dd)Reinstaller\volume\setup.exe; *where D: is the drive letter of the CD drive.*

6.1.1 SOFTWARE INSTALLATION VERIFICATION

- a. Open National Instruments Measurement and Automation Explorer (MAX)
- b. Expand 'My System'
- c. Expand 'Data Neighborhood'
- d. Right-click 'FieldPoint Items' and Open the configuration file 'C:\Program Files\A2000wf\support\FieldPoint.iak'
- e. Expand 'FieldPoint Items'

- f. Right-click on 'a2000wf' and choose 'Go to Comm Resource Configuration'. Verify that the IP network settings agree with the laptop console being used.

6.1.2 ASSIGNING AN IP ADDRESS

You can connect your PC to the A-2000 Auto-Isokinetic Flow Integrator directly using a CAT-5 Ethernet **crossover cable**. You may also communicate with the A-2000 Auto-Isokinetic Flow Integrator by connecting it and your PC to a local area network (LAN) using a standard CAT-5 Ethernet cables. Either way, your A-2000 Auto-Isokinetic Flow must have an IP address assigned to it. You can either attempt to automatically obtain an IP address, known as DYNAMIC IP addressing, from a DHCP server or manually, known as STATIC IP addressing.

You **must** specify a STATIC IP address if your A-2000 Auto-Isokinetic Flow Integrator is not connected to a network and you want to make a direct connection to your A-2000 PC Console. Refer to Section 2.1.3 - Specifying a Static IP Address.

The A-2000 System is shipped with Static IP addressing on both the Auto-Isokinetic Flow Integrator and the laptop console. The default IP Network Settings are:

A-2000 Auto-Isokinetic Flow Integrator

IP Address	192.168.1.4
Submask	255.255.255.0
Gateway	192.168.1.1

A-2000-PC Laptop Console

IP Address	192.168.1.5
Submask	255.255.255.0
Gateway	192.168.1.1

If your A-2000 Auto-Isokinetic Flow Integrator is on a network that has a DHCP server, you may be able to automatically obtain an IP Address from the DHCP server. A DHCP server allocates an IP address to your A-2000 Auto-Isokinetic Flow Integrator each time the A-2000 Auto-Isokinetic Flow Integrator is turned on. You do not need to specify other information such as Subnet Mask if you select the DHCP Server option.

NOTE: If you do not know whether your network has a DHCP server, check with your network administrator for assistance.

To automatically obtain an IP address, select **Obtain IP address from DHCP server**, and then click **Apply**. You must restart the A-2000 Auto-Isokinetic Flow Integrator for any changes to take effect.

Not all DHCP servers are implemented in the same manner. Therefore, some might not be compatible with the A-2000 Auto-Isokinetic Flow Integrator. After you select **Obtain IP Address from DHCP Server** and restart the A-2000 Auto-Isokinetic Flow Integrator, the A-2000 Auto-Isokinetic Flow Integrator tries to obtain an IP address from the DHCP server. If this operation fails, the A-2000 Auto-Isokinetic Flow Integrator automatically restarts and makes a second attempt. After three failed attempts, the A-2000 Auto-Isokinetic Flow Integrator returns to the default configuration with IP address 0.0.0.0. In

this case, you need to explicitly specify the network parameters. Refer to Section 2.1.3 - Specifying a Static IP Address.

In addition, when you use a DHCP server, the server allocates an IP address to the A-2000 Auto-Isokinetic Flow Integrator each time it is turned on. The new IP address might be different than the address previously assigned. If you use the DHCP server to assign an IP address to your A-2000 Auto-Isokinetic Flow Integrator, you need to check the IP address using National Instruments Measurement and Automation Explorer (MAX) each time you power on the A-2000 Auto-Isokinetic Flow Integrator. To avoid needing to check the IP address each time, specify a static IP address for the RT target instead of using a DHCP server.

6.1.3 Specifying a Static IP Address

If you choose to specify an IP address, select **Edit the IP Settings** from the Network Settings tab, fill in the network parameters described below with correct values for your network, then click **Apply**. You must restart the A-2000 Auto-Isokinetic Flow Integrator for any changes to take effect.

IP Address - The unique address of a device on your network. Each IP address is a set of four one to three digit numbers. Each number is in the range from 0 through 255 and is separated by a period. This format is called dotted decimal notation. The IP address 224.102.13.24 is an example of dotted decimal notation.

Subnet Mask - A code that helps the network device determine whether another device is on the same network or a different network. 255.255.255.0 is the most common subnet mask.

Gateway - The IP address of a device that acts as a gateway server, which is a connection between two networks.

DNS Address - The IP address of a network device that stores DNS host names and translates them into IP addresses.

Consult with your network administrator before specifying these parameters. If you do not have a network administrator or you are the network administrator, refer to National Instruments Measurement and Automation Explorer (MAX) IP Advanced help information. If you are uncertain about what to set, click **Suggest Values** to have Measurement and Automation Explorer (MAX) attempt to detect the appropriate settings. Click **Refresh** to revert your changes.

7 APPENDIX

7.1 Isokinetic Equations

$$P_m = P_b + \Delta H/13.6$$

P_m is the pressure used to calculate the equivalent volume of a test meter when calibrating the Venturi flow rate.

P_b is barometric pressure (inHg -or- mmHg).

ΔH is the differential pressure of the Venturi (inH₂O -or- mmH₂O).

$$P_{st} = P_b + \Delta P_{st}/13.6$$

P_{st} is the absolute stack pressure (inHg -or- mmHg).

ΔP_{st} is the differential static pressure (inH₂O -or- mmH₂O).

$$P_v = \text{abs}(P_b - \text{Vac});$$

P_v is the pressure at the head of the Venturi (inHg -or- mmHg).

Vac is the vacuum at the head of the Venturi (inHg -or- mmHg).

$$A_n = \pi D_n^2 / 4 / 144; (\text{for English})$$

A_n is the area of the nozzle (ft²).

π 3.14

D_n is the nozzle diameter (in).

$$A_n = \pi D_n^2 / 4 / 1e6; (\text{for Metric})$$

A_n is the area of the nozzle (m²).

π 3.14

D_n is the nozzle diameter (mm).

$$B_{ws} = B_{ws} 100 / 100;$$

B_{ws} is moisture volume fraction.

100 is percent moisture volume.

$$M_d = (100 - \text{CO}_2 - \text{O}_2) * 0.28 + 0.44 * \text{CO}_2 + 0.32 * \text{O}_2;$$

M_d is dry molecular weight of stack gas.

$$M_{ws} = M_d * (1 - B_{ws}) + 18 * B_{ws};$$

M_{ws} is wet molecular weight of stack gas.

$$Q_v = k * \Delta H^n * \text{sqrt}(T_v / P_v / M_d);$$

Q_v is volumetric flow rate at the Venturi (CFM -or- LPM).

'k' is a calibration constants.

'n' is a calibration constants.

T_v is the temperature at the dryer outlet and Venturi (°R -or- °K).

$$Q_{dsvpm} = Q_v * (T_{std} + 460) * P_v / P_{std} / T_v; (\text{for English});$$

Q_{dsvpm} is flow rate at standard conditions (CFM).

Gas is assumed dry after the dryer.

T_{std} is standard temperature ($^{\circ}F$).

P_{std} is standard pressure (inHg).

$$Q_{dsvpm} = Q_v \cdot (T_{std} + 273.15) \cdot P_v / P_{std} / T_v; \text{ (for Metric)}$$

Q_{dsvpm} is flow rate at standard conditions (LPM).

Gas is assumed dry after the dryer.

T_{std} is standard temperature ($^{\circ}C$).

P_{std} is standard pressure (mmHg).

$$Q_n = Q_v / (1 - B_{ws}) \cdot P_v \cdot T_{st} / T_v / P_{st};$$

Q_n is the flow rate at the nozzle (CFM -or- LPM).

$$Q_m = Q_v / P_m / T_v \cdot T_m \cdot P_v / G_{am};$$

Q_m is the equivalent flow rate of an external test meter (CFM -or- LPM).

G_{am} is the efficiency factor of the test meter.

$$Vol_out = Q_{dsvpm} \cdot t_avg + Vol_in;$$

$$V_{m_out} = Q_m \cdot t_avg + V_{m_in};$$

Vol_out , V_m out, Vol_in , and V_m out are program time integration variables for the time increment t_avg .

$$V_n = Q_n / A_n / 60; \text{ (for English)}$$

V_n is the nozzle velocity (ft/sec).

$$V_n = Q_n / A_n / 6e4; \text{ (for Metric)}$$

V_n is the nozzle velocity (m/sec).

$$V_{st} = 85.49 \cdot C_p \cdot \sqrt{(\Delta Pitot \cdot T_{st} / P_{st} / M_{ws})}; \text{ (for English)}$$

V_{st} is the stack velocity (ft/sec).

$$V_{st} = 34.97 \cdot C_p \cdot \sqrt{(\Delta Pitot \cdot T_{st} / P_{st} / M_{ws})}; \text{ (for Metric)}$$

V_{st} is the stack velocity (m/sec).

$$Iso = 100 \cdot V_n / V_{st};$$

Iso is the percent isokinetic.

7.2 Troubleshooting

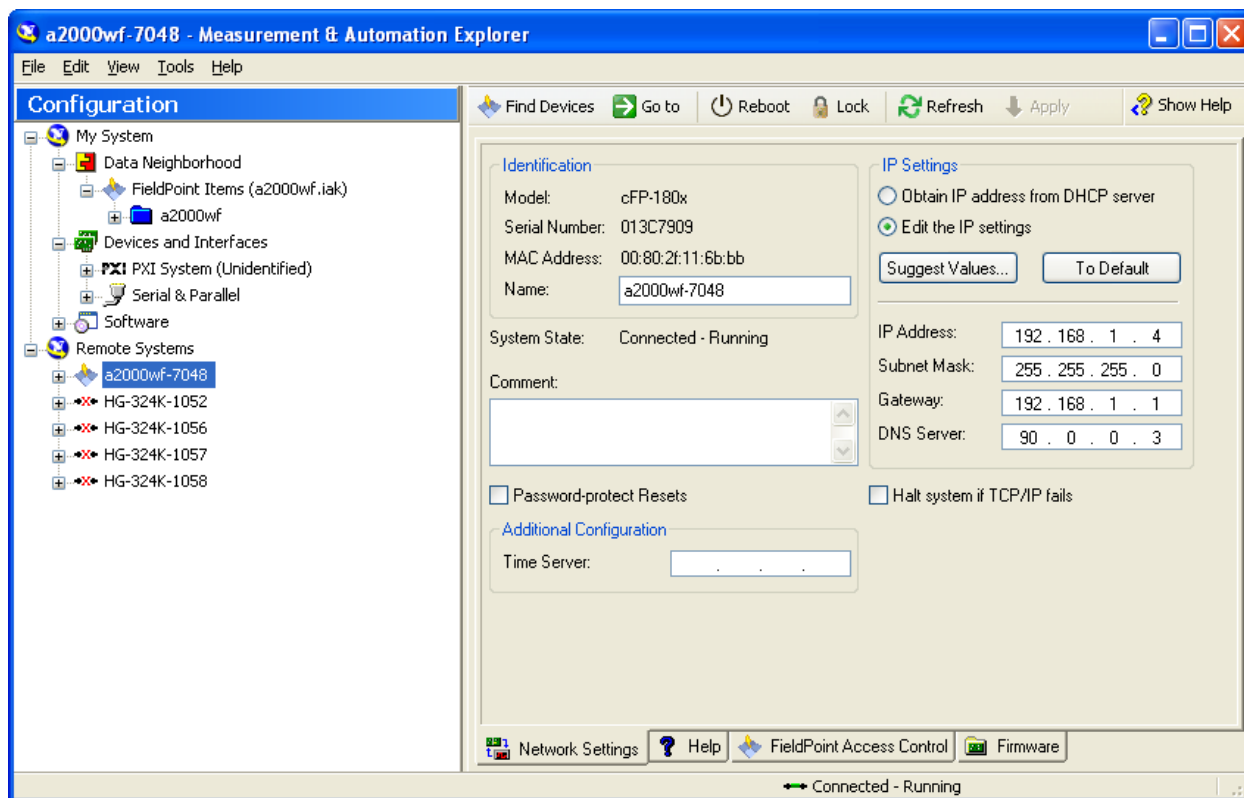
7.2.1 A message appears that the Field Point .iak file can't be found.

Browse or otherwise show the path C:\Program Files\A2000wf\YYYY-MM-DD)\a2000wf.iak, \config\ a2000wf.iak, or \support\FieldPoint.iak.

7.2.2 The program appears to run but the initial values are not correct.

Go to folder properties and set to share. Be sure that files are not read-only. The Vista operating system seems to be very rigid on this requirement.

7.3 Viewing Data in Measurement and Automation Explorer (MAX)

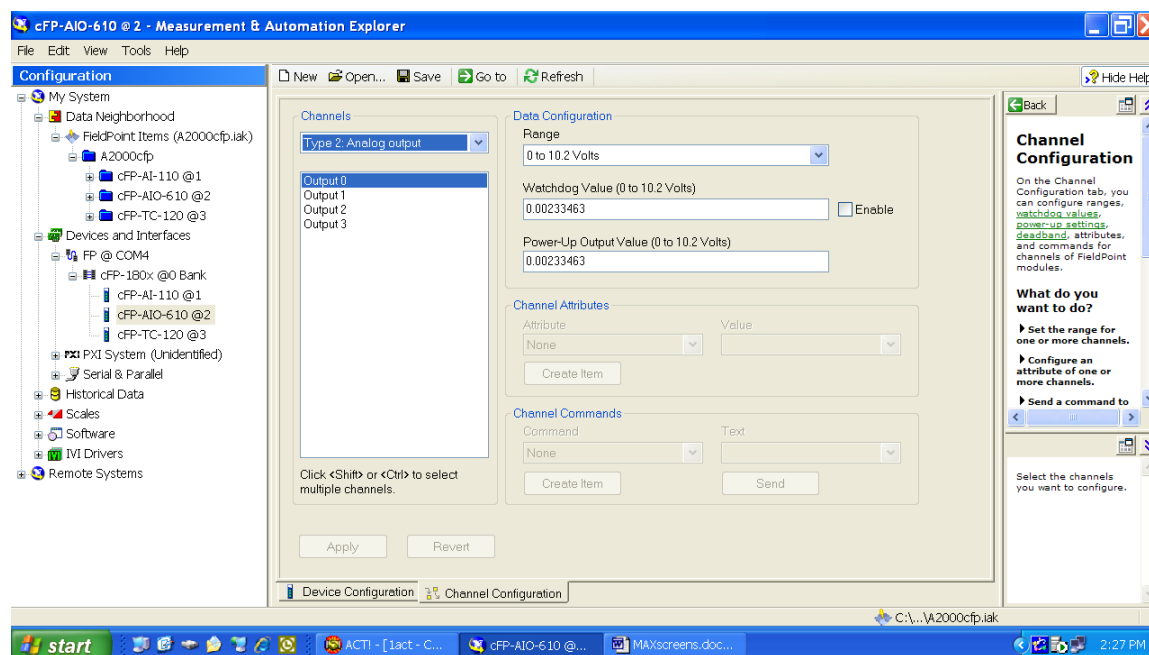
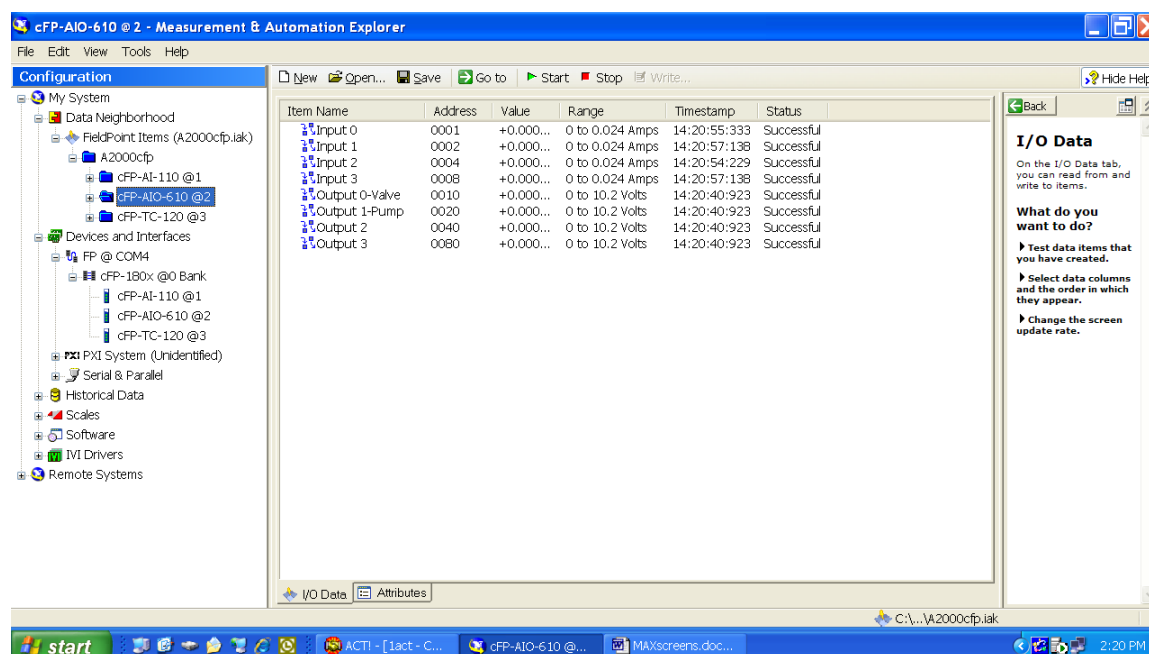


The desktop icon is 'A2000wfxxxx'. This will start the program file 'C:\Program files\A2000wf\ A2000wf1007.exe'. When you first run the A-2000wf program you may be asked to find the configuration file such as 'a2000wf.iak'; browse to the file path and select 'C:\Program Files\A2000wf\config\ a2000wf.iak'. The program should now be fully operational.

Updates for the A-2000 Auto-Isokinetic Flow Integrator software will usually have only the basic executables without the larger files for the Labview Runtime Engine and Field Point. The latest versions of Labview Runtime Engine (LVRunTimeEng.exe) and Field Point (FieldPoint50.exe) can be downloaded from WWW.NI.COM.

Raw data from the data acquisition system may be viewed in Max. The following screens show the data display and the channel configuration. The configuration of the channels is critical for proper operation of the program. Therefore do not change these configurations without having a backup copy of a2000wf.iak. Some programs operate with a project file called FieldPoint.iak. This is usually located in the 'Support' instead of the 'Config' folder. In the latter case it is necessary to open this folder in MAX and set its proper address.

7.5 cFP-AIO-610 Screenshot



7.6 cFP-TC-120 Screenshot

The screenshot shows the 'cFP-TC-120 @ 3 - Measurement & Automation Explorer' window. The left pane displays a tree view of the system hierarchy, including 'My System', 'Data Neighborhood', 'FieldPoint Items (A2000cftp)', 'Devices and Interfaces', and 'Remote Systems'. The main pane shows the 'I/O Data' tab with a table of data points.

Item Name	Address	Value	Range	Timestamp	Status
All	01FF	+13.57...	multi-channel item	14:23:23:970	Open th...
CJ Temperature	0100	+19.70...	-50 to 85 Celsius	14:23:22:892	Successful
Ch0-Stack	0001	+13.57...	-270 to 1770 Celsius	14:23:22:992	Successful
Ch1-Probe	0002	+15.91...	-270 to 1770 Celsius	14:23:23:091	Successful
Ch2-Filter Box	0004	+1770...	-270 to 1770 Celsius	14:23:13:839	Open th...
Ch3-Dryer	0008	+16.63...	-270 to 1770 Celsius	14:23:23:392	Successful
Ch4-Filter Out	0010	+16.44...	-270 to 1770 Celsius	14:23:23:491	Successful
Ch5-DGM-option	0020	+19.15...	-270 to 1770 Celsius	14:23:23:598	Successful
Ch6-aux	0040	+19.08...	-270 to 1770 Celsius	14:23:23:792	Successful
Ch7-Console	0080	+19.36...	-270 to 1770 Celsius	14:23:23:893	Successful

The right pane shows the 'I/O Data' help text, which includes instructions on how to test data items, select data columns, and change the screen update rate.

The screenshot shows the 'cFP-TC-120 @ 3 - Measurement & Automation Explorer' window with the 'Channel Configuration' tab selected. The left pane shows the system hierarchy, and the main pane displays the configuration settings for the channels.

Channels

Type 1: Analog input

Channel 0
Channel 1
Channel 2
Channel 3
Channel 4
Channel 5
Channel 6
Channel 7

Click <Shift> or <Ctrl> to select multiple channels.

Data Configuration

Range: -270 to 1770 Celsius

Watchdog Value: ☐ Enable

Percent Deadband:

Channel Attributes

Attribute: Thermocouple type Value: K

Create Item

Channel Commands

Command: None Text:

Create Item Send

Apply Revert

Device Configuration Channel Configuration

The right pane shows the 'Channel Configuration' help text, which includes instructions on how to set the range, configure attributes, send commands, and set power-up values.

7.7 Summary of Specifications

7.7.1 Accuracy

1. Differential Pressure Transducers -- $\pm 1\%$
2. Barometer -- 0.25%
3. Venturi -- $< 1\%$
4. Data Acquisition – 12 bit resolution
5. Type K thermocouple – 1.5 degree C (0 – 350)